

SKILL SPHERE

Submitted by

BSVIT-21-03

Sawaira Tasleem

&

BSVIT-21-12

Javeria Iram

BSIT (2021-2025)



2026

DEPARTMENT OF INFORMATION TECHNOLOGY

BAHAUDDIN ZAKARIYA UNIVERSITY MULTAN

PAKISTAN

Project Report

Skill Sphere

Submitted by

BSVIT-21-03

Sawera Tasleem

BSVIT-21-12

Javeria Iram

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A REPORT

**SUBMITTED TO THE DEPARTMENT OF INFORMATION TECHNOLOGY
BAHAUDDIN ZAKARIYA UNIVERSITY MULTAN
IN PARTIAL FULFILLMENT OF REQUIREMENTS FOR THE DEGREE OF BS
(IT)
DEPARTMENT OF INFORMATION TECHNOLOGY
BAHAUDDIN ZAKARIYA UNIVERSITY MULTAN PAKISTAN**

Final Approval

This is to clarify that we have read this report submitted by *Sawaira BSVIT-21-03 & Javeria BSVIT-21-12* and it is our judgment that this report is of sufficient standard to warrant its acceptance by Bahauddin Zakariya University, Multan for the degree of BS (IT) (Bachelors of Science in Information Technology).

Committee:

1. External Examiner

Prof. Dr. Maruf Pasha

2. Supervisor

Sir Toseefullah

3. Head of Department

Sir Mohsin Naqi

DEDICATION

To my Loving Parents and Teachers

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor for their continuous guidance, valuable suggestions, and support throughout the development of this project. Their encouragement and constructive feedback played a vital role in shaping this work and improving its overall quality.

I am also thankful to the faculty members of the Department of Information Technology for providing a strong academic foundation and for their assistance whenever needed.

Their knowledge and teaching have been instrumental in completing this project successfully.

I would like to extend my appreciation to my friends and classmates for their cooperation, discussions, and support during different stages of this project. Their input helped me overcome various challenges.

Finally, I am deeply grateful to my family for their constant encouragement, patience, and moral support. Their belief in me gave me the motivation to complete this project with dedication and confidence.

PROJECT BRIEF

PROJECT NAME	Skill Sphere
UNDERTAKEN BY	Sawaira & Javeria
SUPERVISED BY	Toseefullah
STARTING DATE	December 21, 2025
COMPLETION DATE	May 3, 2026
COMPUTER USED	Core i5 ,8GB RAM ,256GB SSD
OPERATING SYSTEM	Windows 10 pro
SOURCE LANGUAGE(S)	React, C-sharp
DBMS USED	SQL Server Management Studio
TOOLS/PACKAGES	MS Office/Visual Studio Code

ABSTRACT

Skill Sphere is a web-based platform designed to facilitate structured learning, assessment evaluation, and collaborative project sharing. The system provides a secure authentication mechanism that allows users to register and access learning environments. Users can enroll in courses, complete lessons, and track their progress automatically. Each course includes quizzes that are auto-graded. Upon successfully passing quizzes, users are required to submit assessments, which are reviewed and approved by an administrator. Only after approval are users eligible to receive a downloadable digital certificate, ensuring credibility and quality assurance. Additionally, the system integrates a project-sharing module where users can upload GitHub project links. These projects are subject to administrative approval before being made publicly accessible.

The admin panel enables efficient management of courses, lessons, assessments, and project approvals. Developed using React, C#, and SQL Server, the system ensures scalability, security, and usability. Overall, the platform aims to provide a comprehensive and controlled learning experience while promoting collaboration and skill development.

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CHAPTER 1

INTRODUCTION

System Introduction

Skill Sphere is a web-based Online Learning and Project Management System designed to provide a structured, self-paced e-learning experience. The platform enables users to enroll in technology-focused courses, complete video-based lessons, attempt auto-graded quizzes, submit written assessments for administrative review, and earn digital certificates upon successful course completion. Additionally, Skill Sphere features a collaborative project sharing system where users can submit their GitHub project links for admin approval and public display.

The system is built using modern web technologies React on the frontend, C# with ASP.NET Core on the backend, and SQL Server as the database management system following client-server architecture. It serves two primary user roles: learners who consume content and progress through courses, and an administrator who manages all course content, reviews submissions, and controls the approval workflow for assessments and projects.

Skill Sphere addresses the growing need for accessible, structured online education platforms that go beyond passive video watching by enforcing a controlled learning progression: users must complete all lessons, pass a quiz with a minimum threshold, and receive admin approval on their assessment before a certificate is generated. This ensures that certification carries genuine value and reflects verified learning outcomes.

Background of the System

The rapid growth of online education has transformed how individuals acquire new skills, particularly in the field of information technology. Platforms such as Coursera, Udemy, and Khan Academy have demonstrated the demand for structured, self-paced learning environments. However, many existing platforms either lack proper assessment validation mechanisms or rely entirely on automated evaluation without any human oversight, which can compromise the credibility of issued certificates.

In the context of academic institutions and small-scale learning communities, there is a need for a lightweight yet robust e-learning system that combines automated progress

tracking with manual quality assurance. Most open-source or small-scale LMS solutions do not offer a built-in project sharing ecosystem or enforce a multi-step completion workflow that includes admin-reviewed assessments as a prerequisite for certification.

Skill Sphere was conceived to fill this gap. It provides a complete learning lifecycle — from course enrollment and lesson completion to quiz evaluation, assessment review, certificate generation, and project showcasing — within a single integrated platform. The system is designed as a Final Year Project (FYP) for the Bachelor of Science in Information Technology (BSIT) program at Government Graduate College Vehari, affiliated with Bahauddin Zakariya University Multan, under the supervision of Sir Toseefullah.

The project draws inspiration from established e-learning platforms while introducing a controlled progression model where every stage of learning is tracked and validated before certification, ensuring that the learning outcomes are meaningful and verifiable.

Objectives of the System

The primary objectives of Skill Sphere are as follows:

1. To develop a secure and user-friendly web-based learning management system that allows users to register, log in, and manage their learning profiles.
2. To provide structured course content delivery through video-based lessons with automatic progress tracking, ensuring users can monitor their advancement through each course.
3. To implement an automated quiz evaluation system with a defined passing criterion (minimum 7 out of 10 correct answers), enforcing a minimum knowledge threshold before users can proceed further.
4. To incorporate a written assessment submission and admin approval workflow, adding a layer of human review to validate the quality of user learning before certificate issuance.
5. To enable automatic generation of downloadable digital certificates upon successful completion of all course requirements — including lesson completion, quiz passing, and assessment approval.
6. To build a project sharing ecosystem where users can submit GitHub project links for admin review, and approved projects become publicly accessible to all platform users, fostering collaboration and knowledge sharing.
7. To develop a comprehensive admin panel that provides full control over course management, lesson management, assessment review, project approval, and user activity monitoring.
8. To ensure the system follows security best practices including password hashing, role-based access control, and protection against unauthorized access.

Scope of the System

Skill Sphere is a web-based e-learning and project management platform that covers the complete learning lifecycle from registration to certification. On the user side, the system allows secure registration and login, course browsing and enrollment, structured lesson completion with automatic progress tracking, auto-graded MCQ quizzes, written assessment submission for admin review, and downloadable digital certificate generation upon meeting all completion criteria. Users can also submit their GitHub project links for admin approval, and approved projects become publicly accessible to all platform users. On the admin side, the system provides a dedicated panel for managing courses and lessons, reviewing and approving or rejecting assessments and project submissions, and monitoring overall user activity. The system is built using React (Vite) on the frontend, C# with ASP.NET Core Web API on the backend, and SQL Server (SSMS) as the database, following a client-server architecture with RESTful API communication. The detailed breakdown of all functional capabilities is covered in the Functional Requirements chapter of this document.

The system does not include payment or subscription modules, file-based project uploads (only GitHub links), multi-admin role hierarchy, real-time communication features, or mobile application support. The quiz format is fixed at 10 MCQs per course, and only a single admin account is supported.

Purpose of the System

The purpose of Skill Sphere is to provide an accessible, structured, and quality-controlled online learning platform that bridges the gap between passive content consumption and verified skill acquisition.

Many existing e-learning solutions allow users to watch videos and receive certificates with minimal or no validation of actual learning. Skill Sphere addresses this by implementing a multi-step completion workflow: users must not only watch all course lessons but also pass an auto-graded quiz and receive admin approval on a written assessment before a certificate is issued. This ensures that every certificate generated by the platform represents a genuine and validated learning achievement.

Beyond individual learning, the platform also serves as a project showcase hub. By allowing users to submit their GitHub projects for admin review and public display, Skill Sphere encourages practical application of learned skills and creates a community-driven portfolio space where users can discover and learn from each other's work.

The system is developed as a Final Year Project to demonstrate proficiency in full-stack web development, database design, RESTful API architecture, and software engineering principles — fulfilling the academic requirements of the BSIT program (2021-2025) at Government Graduate College Vehari.

CHAPTER 2

SYSTEM ANALYSIS

2.1 Feasibility Study

A feasibility study is conducted to evaluate whether the proposed system is practical, achievable, and worth developing. The feasibility of Skill Sphere has been analyzed from three perspectives: technical feasibility, economic feasibility, and operational feasibility.

2.1.1 Technical Feasibility

Technical feasibility assesses whether the required technologies, tools, and expertise are available to successfully develop and deploy the system. Skill Sphere is built using well-established and widely supported technologies. The frontend is developed in React using Vite as the build tool, which provides a fast and modern development environment with extensive community support and documentation. The backend is powered by C# with ASP.NET Core Web API, a mature and enterprise-grade framework that offers built-in support for RESTful API development, authentication, and middleware configuration. SQL Server, managed through SQL Server Management Studio (SSMS), serves as the relational database, providing reliable data storage, querying, and management capabilities.

All the tools and technologies used in this project are freely available or included in academic/community editions. Visual Studio and Visual Studio Code serve as the primary development environments, and Git is used for version control. The development team possesses the required knowledge and skills in full-stack web development gained through coursework and practical experience during the BSIT program. No specialized hardware or proprietary software is required, making the project fully feasible from a technical standpoint.

2.1.2 Economic Feasibility

Economic feasibility determines whether the project can be completed within a reasonable budget. Since Skill Sphere is developed as an academic Final Year Project, the cost considerations are minimal. All technologies and frameworks used in the development are open-source or available under free academic licenses. React, Vite, ASP.NET Core, and SQL Server Express/Developer Edition are all available at no cost. The development is carried out on personal computers already owned by the team members, eliminating any hardware procurement costs.

There are no hosting or deployment expenses during the development and testing phase, as the system runs on localhost using local database instances. Should the system be deployed in the future, affordable cloud hosting options such as Azure (with free-tier

student plans) or other budget-friendly providers can be utilized. No third-party paid APIs or services are integrated into the system. Therefore, the project is economically feasible with virtually zero financial investment required for development and testing.

2.1.3 Operational Feasibility

Operational feasibility evaluates whether the system will be effectively used by its intended users once deployed. Skill Sphere is designed with simplicity and ease of use as core priorities. The user interface follows a clean and intuitive layout, allowing learners to browse courses, track their progress, attempt quizzes, and submit assessments without requiring any technical expertise. The admin panel provides straightforward controls for managing courses, reviewing submissions, and monitoring user activity.

The system automates key operations such as progress tracking, quiz grading, and certificate generation, reducing the manual workload for both users and administrators. The controlled progression model where users must complete lessons, pass quizzes, and receive assessment approval before certification is logically structured and easy to understand. Users familiar with any basic e-learning platform will find Skill Sphere's workflow intuitive and self-explanatory. The single-admin model keeps the operational overhead minimal, making it practical for small-scale deployment in academic or training environments.

2.2 Software Requirement Specification

This section defines the detailed functional requirements, non-functional requirements, and expected deliverables of the Skill Sphere system.

2.2.1 Functional Requirements

Functional requirements describe the specific behaviors, features, and functions that the system must perform. The following subsections detail the functional requirements organized by system module.

2.2.1.1 Authentication and User Management

The system shall provide a secure registration page where users can create an account by providing their name, email address, and password. The system shall validate all input fields during registration and display appropriate error messages for invalid or incomplete entries. The system shall hash all passwords before storing them in the database to ensure security. The system shall provide a login page where registered users can authenticate using their email and password. The system shall provide a Forgot Password feature that allows users to reset their password by verifying their registered email address. When a user requests a password reset, the system shall validate that the email exists in the database, allow the user to set a new password, hash the new password, and update the stored password accordingly. The system shall implement role-

based access control to distinguish between regular users and the admin account. The system shall maintain a single admin account with elevated privileges for content and submission management. The system shall prevent unauthorized users from accessing admin-only routes and functionalities.

2.2.1.2 Course Management

The admin shall be able to create new courses by providing a course title, description, and relevant details. The admin shall be able to update existing course information or delete courses from the system. The system shall display all available courses on a courses page accessible to registered users. Each course shall contain a structured set of video-based lessons organized in a sequential order. The admin shall be able to add, update, or remove lessons within any course.

2.2.1.3 Course Enrollment

The system shall allow registered users to enroll in any available course. The system shall store enrollment records in the database, linking each user to their enrolled courses. The system shall display enrolled courses on the user's dashboard with current progress status.

2.2.1.4 Lesson Completion and Progress Tracking

Each course shall contain multiple lessons delivered through embedded YouTube video content. The system shall provide a side panel for lesson navigation, allowing users to move between lessons within a course. The system shall automatically track lesson completion as users watch the video content. The system shall calculate and display the progress percentage for each enrolled course based on the number of completed lessons out of total lessons. The system shall update progress in real-time as users complete lessons.

2.2.1.5 Quiz System

Each course shall have an associated quiz containing 10 multiple-choice questions (MCQs). The system shall auto-grade quizzes immediately upon submission and display the score to the user. The passing criterion shall be a minimum of 7 correct answers out of 10. If the user scores 7 or above, the system shall unlock the assessment submission for that course. If the user scores below 7, the system shall require the user to reattempt the quiz. The quiz button shall remain visible at all times with appropriate lock or unlock states based on lesson completion status.

2.2.1.6 Assessment Submission and Review

After passing the quiz, the system shall allow users to submit a written assessment for the corresponding course. The system shall store the assessment submission with a status field that can be Pending, Approved, or Rejected. The admin shall be able to view all pending assessment submissions from the admin panel. The admin shall be able to approve or reject each assessment submission. The system shall notify the user of the assessment status on their dashboard. The assessment button shall remain visible at all times with appropriate lock or unlock states based on quiz completion status.

2.2.1.7 Certificate Generation

The system shall generate a digital certificate only when all three conditions are met: all lessons in the course are completed, the quiz is passed with a score of 7 or above, and the assessment is approved by the admin. The certificate shall contain the user's name, the course name, and the completion status. The system shall provide the certificate as a downloadable file. The certificate button shall remain visible at all times with appropriate lock or unlock states based on assessment approval status. The system shall automatically generate the certificate upon admin approval of the assessment without requiring any additional user action.

2.2.1.8 Project Sharing System

The system shall allow users to upload GitHub project links through a project submission form. The system shall not accept file uploads; only valid GitHub URLs shall be accepted. Submitted projects shall not be immediately visible to other users. The admin shall review each project submission and either approve or reject it. Approved projects shall appear in a public project section accessible to all registered users. Rejected projects shall not be displayed publicly. Users shall be able to browse and access approved project links submitted by other users.

2.2.1.9 Admin Panel

The admin shall have access to a dedicated admin dashboard separate from the regular user dashboard. The admin panel shall provide the ability to add, update, and delete courses. The admin panel shall provide the ability to add and manage lessons within courses. The admin panel shall display all pending assessment submissions for review and allow approval or rejection. The admin panel shall display all pending project submissions for review and allow approval or rejection. The admin panel shall provide an overview of user activities and system statistics.

2.2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes, constraints, and standards that the system must adhere to beyond its core functionality.

2.2.2.1 Security

The system shall hash all user passwords using a secure hashing algorithm before storing them in the database. The system shall implement role-based access control to ensure that admin functionalities are not accessible to regular users. The system shall protect all API endpoints against unauthorized access through proper authentication and authorization middleware. The system shall validate and sanitize all user inputs to prevent SQL injection, cross-site scripting (XSS), and other common security vulnerabilities.

2.2.2.2 Performance

The system shall auto-grade quizzes and return results to the user in real-time without noticeable delay. The system shall load course pages, dashboards, and lesson content within a reasonable response time under normal operating conditions. The database queries shall be optimized to ensure efficient retrieval and storage of user progress, quiz results, and submission records.

2.2.2.3 Usability

The system shall provide a clean, simple, and intuitive user interface that requires no prior training to use. The navigation structure shall be logical and consistent across all pages, allowing users to easily access courses, quizzes, assessments, and projects. The system shall display clear feedback messages for all user actions including successful submissions, errors, validation failures, and status updates. The system shall use a consistent visual theme (teal and coral color scheme) throughout the application.

2.2.2.4 Reliability

The system shall accurately track and store lesson progress, quiz scores, assessment statuses, and project approval statuses without data loss or inconsistency. The system shall maintain data integrity across all database tables through proper use of foreign keys, constraints, and transactional operations. The system shall handle errors gracefully and display user-friendly error messages instead of exposing raw system errors.

2.2.2.5 Scalability

The system architecture shall support the addition of new courses, lessons, quizzes, and users without requiring structural changes to the codebase or database schema. The client-server architecture with RESTful APIs shall allow the frontend and backend to be

scaled independently if needed in the future. The database design shall accommodate growing data volumes without significant performance degradation.

2.2.2.6 Maintainability

The codebase shall follow clean coding practices with proper separation of concerns between frontend components and backend controllers. The backend shall be organized into clearly defined controllers (Auth, Admin, Courses, Enrollments, LessonProgress, Quiz, Assessment, Certificate, Projects, UserProjects) for easy maintenance and future modifications. The frontend shall use modular React components that can be updated or replaced independently.

2.2.2.7 Compatibility

The system shall be compatible with modern web browsers including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. The frontend shall be responsive enough to function on different screen sizes, though the primary target is desktop and laptop browsers.

2.2.3 Deliverables

The following deliverables shall be produced as part of the Skill Sphere project:

1. A fully functional web-based Online Learning and Project Management System with all features described in the functional requirements, including user authentication, course management, lesson tracking, quiz evaluation, assessment review, certificate generation, and project sharing.
2. A comprehensive Software Requirement Specification (SRS) document covering the system introduction, feasibility study, functional and non-functional requirements, system design, database design, and all supporting documentation.
3. A relational database (SkillSphereDB) implemented in SQL Server containing all necessary tables for users, courses, lessons, enrollments, lesson progress, quizzes, quiz questions, quiz attempts, assessment submissions, certificates, projects, and user projects.
4. A RESTful API backend built with ASP.NET Core C# providing all endpoints required for frontend-backend communication, including controllers for authentication, course management, enrollment, progress tracking, quiz handling, assessment processing, certificate generation, and project management.
5. A React-based frontend application built with Vite, providing all user-facing pages including Landing Page, Authentication Pages, User Dashboard, Admin Dashboard, Course Pages, Quiz Pages, Assessment Pages, Certificate Pages, and Project Pages.
6. A project demonstration and presentation showcasing the complete working system to the project supervisor and evaluation panel.

7. Source code repository with version-controlled codebase maintained throughout the development lifecycle.

CHAPTER 3:

SYSTEM DESIGN

3.1 Introduction to System Design

System design is the process of defining the architecture, components, modules, interfaces, and data flow of a system to satisfy the specified requirements. It translates the functional and non-functional requirements identified in the previous chapter into a structured blueprint that guides the actual development process.

The system design for Skill Sphere follows a client-server architecture where the React frontend communicates with the ASP.NET Core C# backend through RESTful API endpoints, and the backend interacts with a SQL Server database for data storage and retrieval. The design is modeled using Unified Modeling Language (UML) diagrams to visually represent the system's structure, behavior, and interactions. This chapter covers use case modeling, activity diagrams, sequence diagrams, class diagrams, data flow diagrams, database design, system architecture, and the overall system workflow.

3.2 System Design using UML

Unified Modeling Language (UML) is a standardized visual modeling language used to specify, visualize, and document the design of a software system. The following subsections present the UML models for Skill Sphere.

3.2.1 Use Cases

Use Case 1: User Registration

Field	Description
Use Case ID	UC-01
Use Case Name	User Registration
Actor	User
Description	A new user creates an account on the Skill Sphere platform by providing their name, email, and password.
Preconditions	The user is not already registered with the provided email address.
Main Flow	1. User navigates to the registration page.

	- User enters name, email, and password. - System validates input fields. - System checks if the email is already registered. - System hashes the password and stores user data in the database. - System redirects user to the login page with a success message.
Alternative Flow	If the email is already registered, the system displays an error message and prompts the user to use a different email or log in. If input validation fails, the system highlights the invalid fields with error messages.
Postconditions	A new user account is created and stored in the database.

Use Case 2: User Login

Field	Description
Use Case ID	UC-02
Use Case Name	User Login
Actor	User, Admin
Description	A registered user or admin logs into the system using their email and password.
Preconditions	The user must have a registered account.
Main Flow	- User navigates to the login page. - User enters email and password. - System validates credentials against the database. - System checks the user role. - If the role is admin, the system redirects to the Admin Dashboard. - If the role is user, the system redirects to the User Dashboard.
Alternative Flow	If credentials are invalid, the system displays an error message and prompts the user to try again.
Postconditions	The user is authenticated and redirected to the appropriate dashboard based on their role.

Use Case 3: Forgot Password

Field	Description
Use Case ID	UC-03
Use Case Name	Forgot Password
Actor	User
Description	A user who has forgotten their password can reset it by verifying their registered email address and setting a new password.
Preconditions	The user must have a registered account with a valid email address in the system.
Main Flow	- User clicks the "Forgot Password" link on the login page. - System displays the password reset form. - User enters their registered email address. - System validates that the email exists in the database. - System prompts the user to enter a new password and confirm it. - System validates the new password against password requirements. - System hashes the new password and updates it in the database. - System displays a success message and redirects the user to the login page.

Alternative Flow	If the email does not exist in the database, the system displays an error message informing the user that no account is associated with the entered email. If the new password and confirmation do not match, the system displays an error and prompts the user to re-enter.
Postconditions	The user's password is updated in the database, and they can log in using the new password.

Use Case 4: Browse Courses

Field	Description
Use Case ID	UC-04
Use Case Name	Browse Courses
Actor	User
Description	A registered user browses the list of available courses on the platform.
Preconditions	The user must be logged in.
Main Flow	1. User navigates to the Courses page. 2. System retrieves all available courses from the database. 3. System displays the course list with titles and descriptions.
Alternative Flow	If no courses are available, the system displays a message indicating no courses are currently offered.
Postconditions	The user views the list of available courses.

Use Case 5: Enroll in Course

Field	Description
Use Case ID	UC-05
Use Case Name	Enroll in Course
Actor	User
Description	A user enrolls in a selected course to begin learning.
Preconditions	The user must be logged in. The user must not already be enrolled in the selected course.
Main Flow	1. User selects a course from the course list. 2. User clicks the Enroll button. 3. System creates an enrollment record linking the user to the course. 4. System redirects the user to the course lessons page.
Alternative Flow	If the user is already enrolled, the system displays the course directly without creating a duplicate enrollment.
Postconditions	An enrollment record is created in the database, and the course appears on the user's dashboard.

Use Case 6: Complete Lessons

Field	Description
Use Case ID	UC-06
Use Case Name	Complete Lessons

Actor	User
Description	A user watches video lessons within an enrolled course, and the system tracks completion progress.
Preconditions	The user must be enrolled in the course.
Main Flow	1. User opens an enrolled course. 2. System displays the lesson list in a side panel. 3. User selects a lesson. 4. System loads the embedded YouTube video for the selected lesson. 5. System automatically marks the lesson as completed after the user watches the video. 6. System updates the overall course progress percentage.
Alternative Flow	If the user has already completed a lesson, the system marks it as completed in the side panel and does not re-track progress.
Postconditions	Lesson completion is recorded, and the course progress percentage is updated in the database.

Use Case 7: Attempt Quiz

Field	Description
Use Case ID	UC-07
Use Case Name	Attempt Quiz
Actor	User
Description	A user attempts the MCQ quiz associated with a course. The system auto-grades the quiz and determines whether the user can proceed.
Preconditions	The user must have completed all lessons in the course.
Main Flow	1. User clicks the Quiz button on the course page. 2. System loads 10 MCQs for the course. 3. User answers all questions and submits the quiz. 4. System auto-grades the quiz and calculates the score. 5. If the score is 7 or above, the system marks the quiz as passed and unlocks the assessment. 6. System displays the score and result to the user.
Alternative Flow	If the score is below 7, the system displays the score and informs the user that they must reattempt the quiz. The assessment remains locked.
Postconditions	The quiz attempt and score are stored in the database. If passed, the assessment submission is unlocked.

Use Case 8: Submit Assessment

Field	Description
Use Case ID	UC-08
Use Case Name	Submit Assessment
Actor	User
Description	After passing the quiz, the user submits a written assessment for admin review.
Preconditions	The user must have passed the quiz with a score of 7 or above.
Main Flow	1. User clicks the Assessment button on the course page. 2. System displays the assessment submission form.

	3. User writes and submits the assessment. 4. System stores the submission with a status of Pending. 5. System confirms successful submission to the user.
Alternative Flow	If the quiz has not been passed, the assessment button remains locked and the user cannot access the submission form.
Postconditions	The assessment submission is stored in the database with Pending status, awaiting admin review.

Use Case 9: Review Assessment (Admin)

Field	Description
Use Case ID	UC-09
Use Case Name	Review Assessment
Actor	Admin
Description	The admin reviews a pending assessment submission and either approves or rejects it.
Preconditions	The admin must be logged in. There must be at least one pending assessment submission.
Main Flow	1. Admin navigates to the assessment review section in the Admin Panel. 2. System displays all pending assessment submissions. 3. Admin selects a submission and reviews the content. 4. Admin clicks Approve or Reject. 5. System updates the assessment status in the database.
Alternative Flow	If no pending assessments exist, the system displays a message indicating there are no submissions to review.
Postconditions	The assessment status is updated to Approved or Rejected in the database. If approved, the certificate generation is unlocked for the user.

Use Case 10: Download Certificate

Field	Description
Use Case ID	UC-10
Use Case Name	Download Certificate
Actor	User
Description	The user downloads a digital certificate after completing all course requirements.
Preconditions	All lessons must be completed. The quiz must be passed. The assessment must be approved by admin.
Main Flow	1. System detects that all three conditions are met. 2. System automatically generates the certificate containing the user's name, course name, and completion status. 3. User clicks the Certificate button. 4. System provides the certificate as a downloadable file.
Alternative Flow	If any of the three conditions are not met, the certificate button remains locked and the user cannot access the certificate.
Postconditions	The certificate record is stored in the database, and the user downloads the certificate file.

Use Case 11: Submit Project Link

Field	Description
Use Case ID	UC-11
Use Case Name	Submit Project Link
Actor	User
Description	A user submits a GitHub project link for admin review.
Preconditions	The user must be logged in.
Main Flow	1. User navigates to the Upload Project page. 2. User enters the project title, description, and GitHub link. 3. System validates that the link is a valid GitHub URL. 4. System stores the submission with a Pending status. 5. System confirms successful submission to the user.
Alternative Flow	If the URL is not a valid GitHub link, the system displays an error message and prompts the user to enter a correct link.
Postconditions	The project submission is stored in the database with Pending status, awaiting admin review.

Use Case 12: Review Project Submission (Admin)

Field	Description
Use Case ID	UC-12
Use Case Name	Review Project Submission
Actor	Admin
Description	The admin reviews a pending project submission and either approves or rejects it.
Preconditions	The admin must be logged in. There must be at least one pending project submission.
Main Flow	1. Admin navigates to the project review section in the Admin Panel. 2. System displays all pending project submissions. 3. Admin selects a submission and reviews the project details and GitHub link. 4. Admin clicks Approves or Reject. 5. System updates the project status in the database.
Alternative Flow	If no pending projects exist, the system displays a message indicating there are no submissions to review.
Postconditions	The project status is updated to Approved or Rejected. Approved projects become visible in the public project section.

Use Case 13: Browse Approved Projects

Field	Description
Use Case ID	UC-13
Use Case Name	Browse Approved Projects
Actor	User
Description	A user browses and accesses publicly approved projects shared by other users.
Preconditions	The user must be logged in.
Main Flow	1. User navigates to the Projects page. 2. System retrieves all approved projects from the database.

	3. System displays the project titles, descriptions, and GitHub links. 4. User clicks on a project link to view the GitHub repository.
Alternative Flow	If no approved projects exist, the system displays a message indicating no projects are currently available.
Postconditions	The user views and accesses approved project links.

Use Case 14: Manage Courses (Admin)

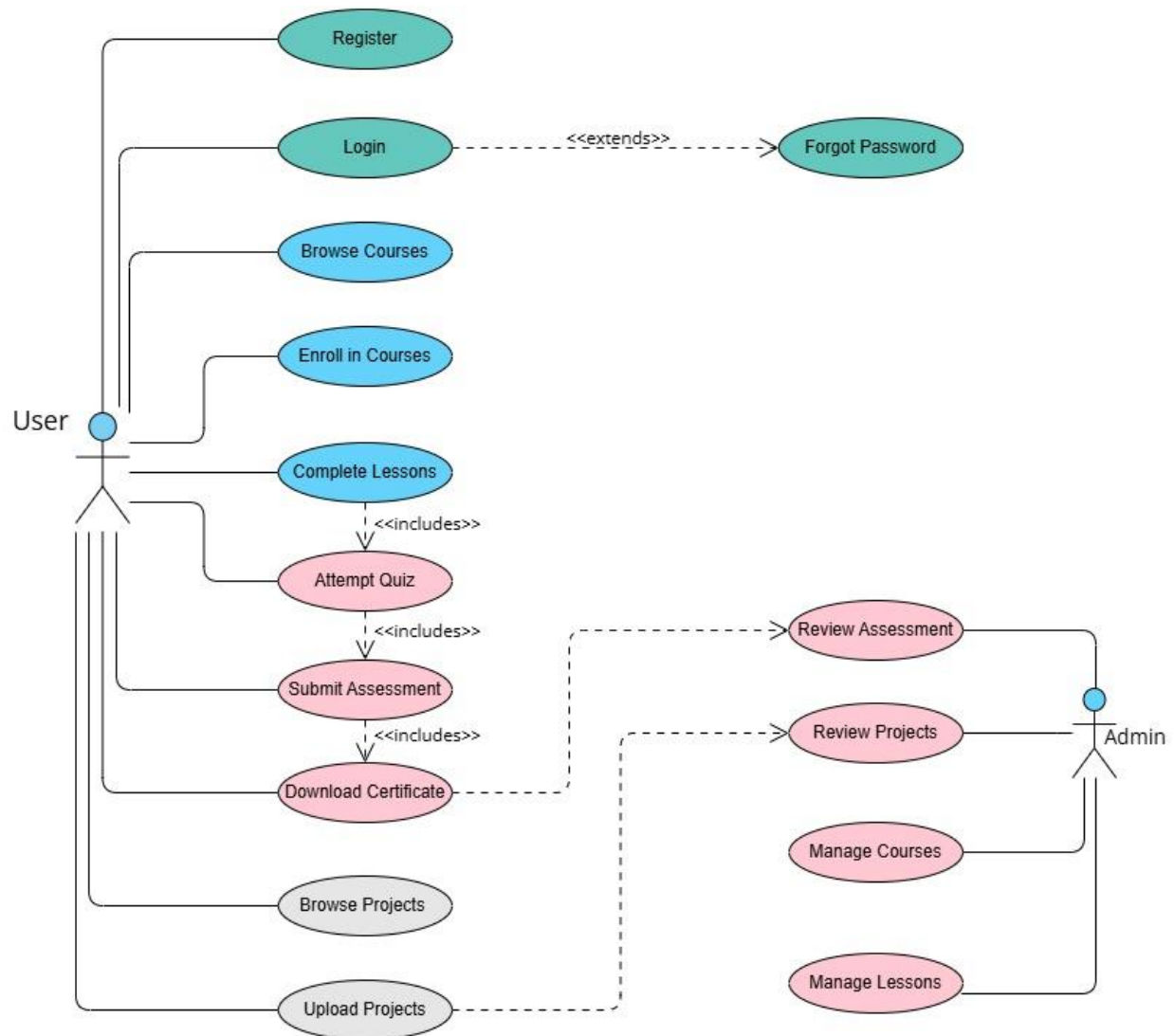
Field	Description
Use Case ID	UC-14
Use Case Name	Manage Courses
Actor	Admin
Description	The admin creates, updates, or deletes courses in the system.
Preconditions	The admin must be logged in.
Main Flow	1. Admin navigates to the course management section in the Admin Panel. 2. To add a course: Admin enters course title, description, and other details, then clicks Save. 3. To update a course: Admin selects an existing course, modifies the details, and clicks Update. 4. To delete a course: Admin selects a course and clicks Delete. 5. System performs the requested operation and updates the database.
Alternative Flow	If the admin attempts to delete a course with active enrollments, the system may display a warning before proceeding.
Postconditions	The course data in the database is created, updated, or deleted as per the admin's action.

Use Case 15: Manage Lessons (Admin)

Field	Description
Use Case ID	UC-15
Use Case Name	Manage Lessons
Actor	Admin
Description	The admin adds, updates, or removes lessons within a course.
Preconditions	The admin must be logged in. The course must already exist.
Main Flow	1. Admin selects a course from the admin panel. 2. Admin navigates to the lessons section of the selected course. 3. To add a lesson: Admin enters lesson title and YouTube video URL, then clicks Save. 4. To update a lesson: Admin modifies the lesson details and clicks Update. 5. To remove a lesson: Admin selects a lesson and clicks Delete. 6. System performs the operation and updates the database.
Alternative Flow	If an invalid YouTube URL is provided, the system displays an error and prompts for correction.
Postconditions	The lesson data within the selected course is created, updated, or deleted as per the admin's action.

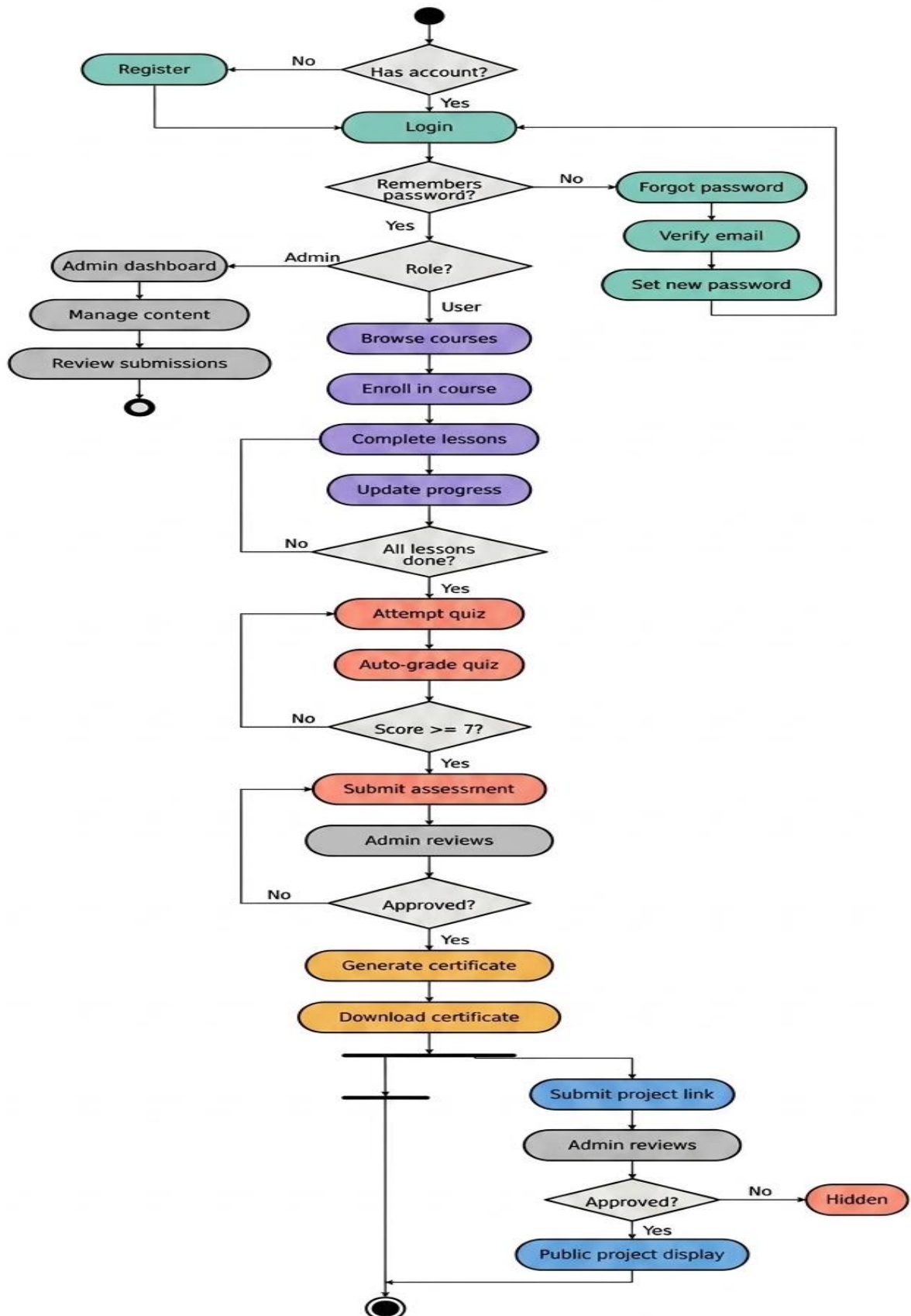
3.2.2 Use Case Diagram

The Use Case Diagram provides a high-level visual overview of the system's functionality and the interactions between actors and use cases. Skill Sphere has two actors: User and Admin.



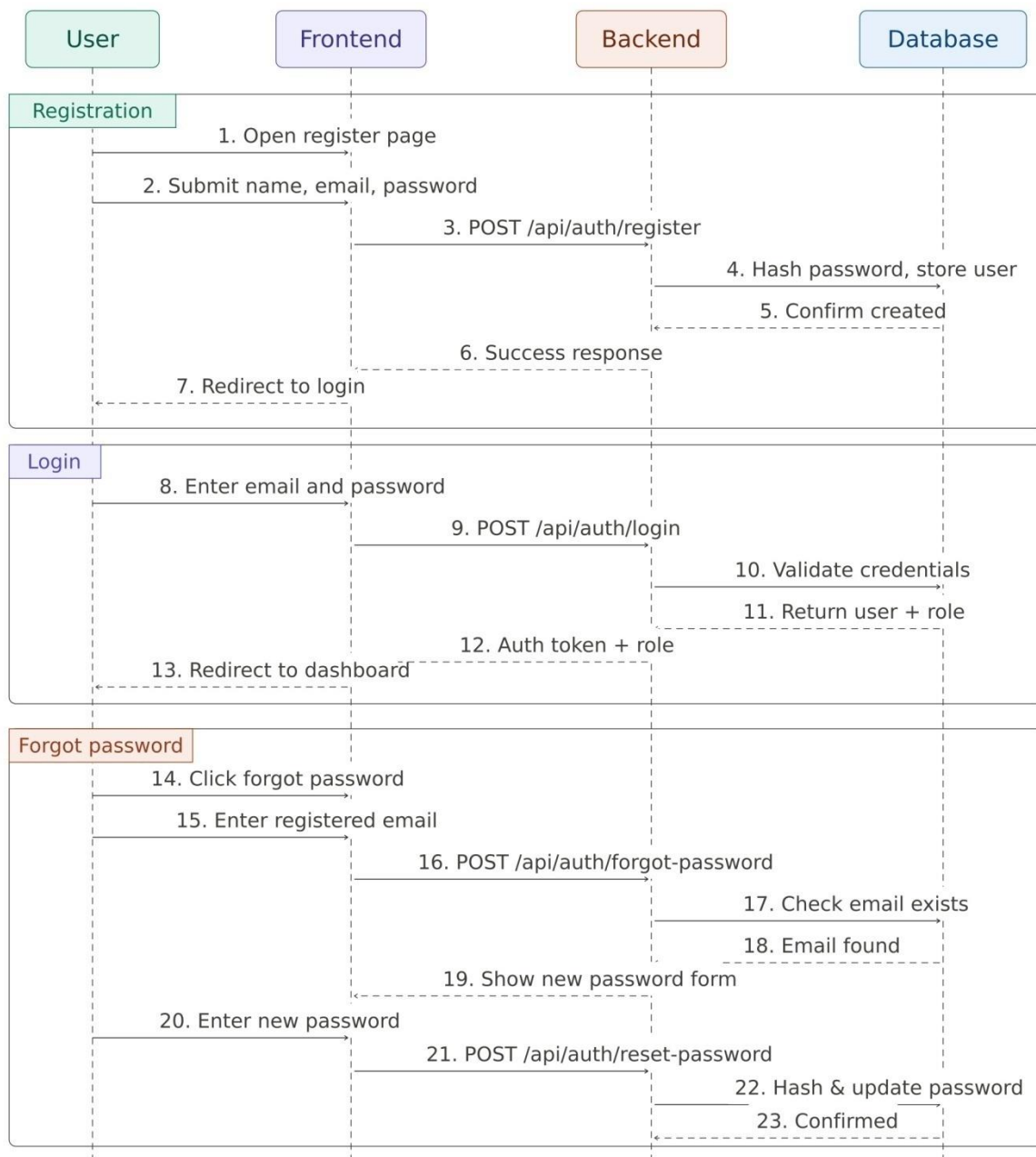
3.2.3 Activity Diagram

The Activity Diagram illustrates the flow of activities within the system from the user's perspective, showing the sequence of actions, decision points, and parallel activities.

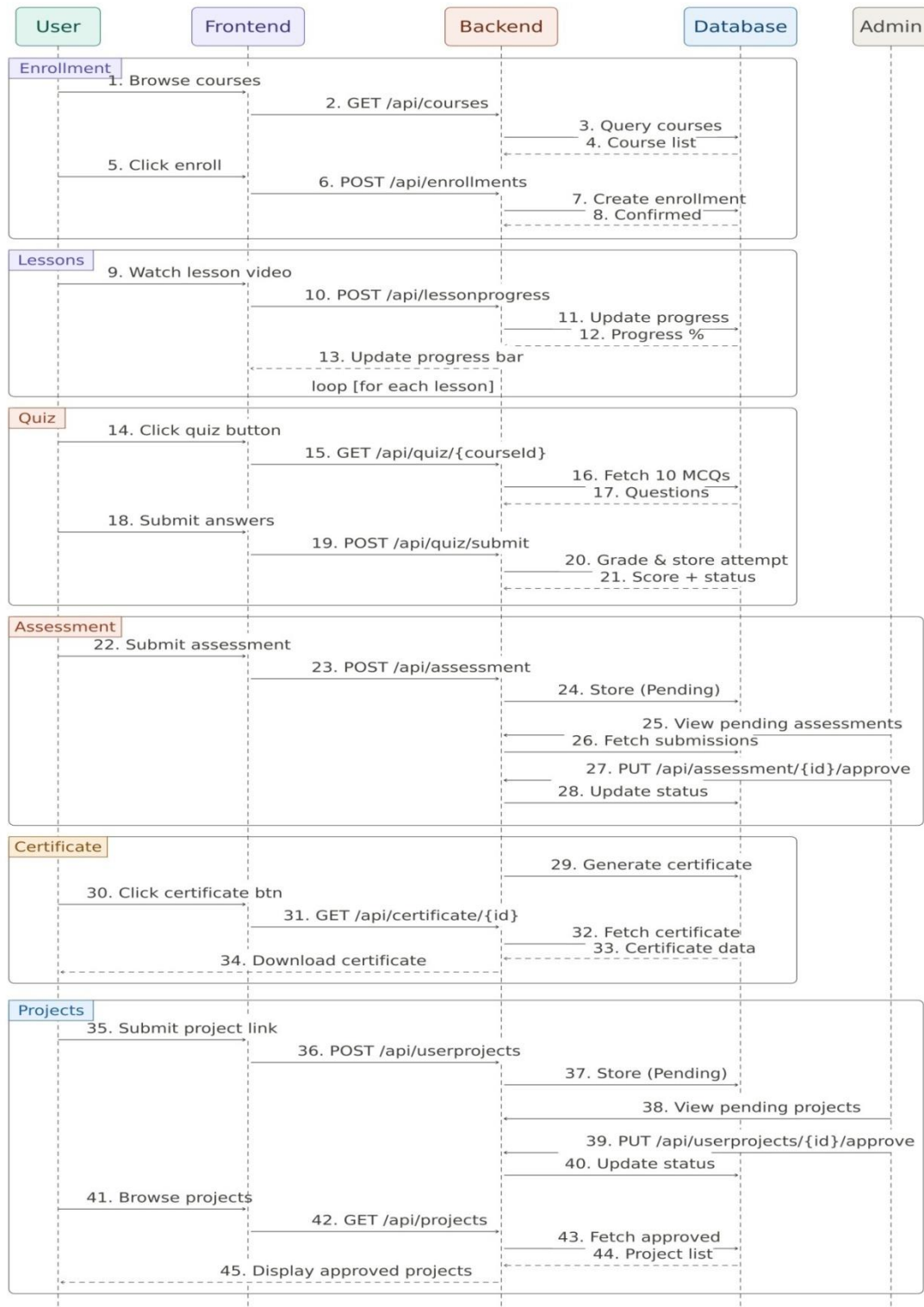


3.2.4 Sequence Diagram

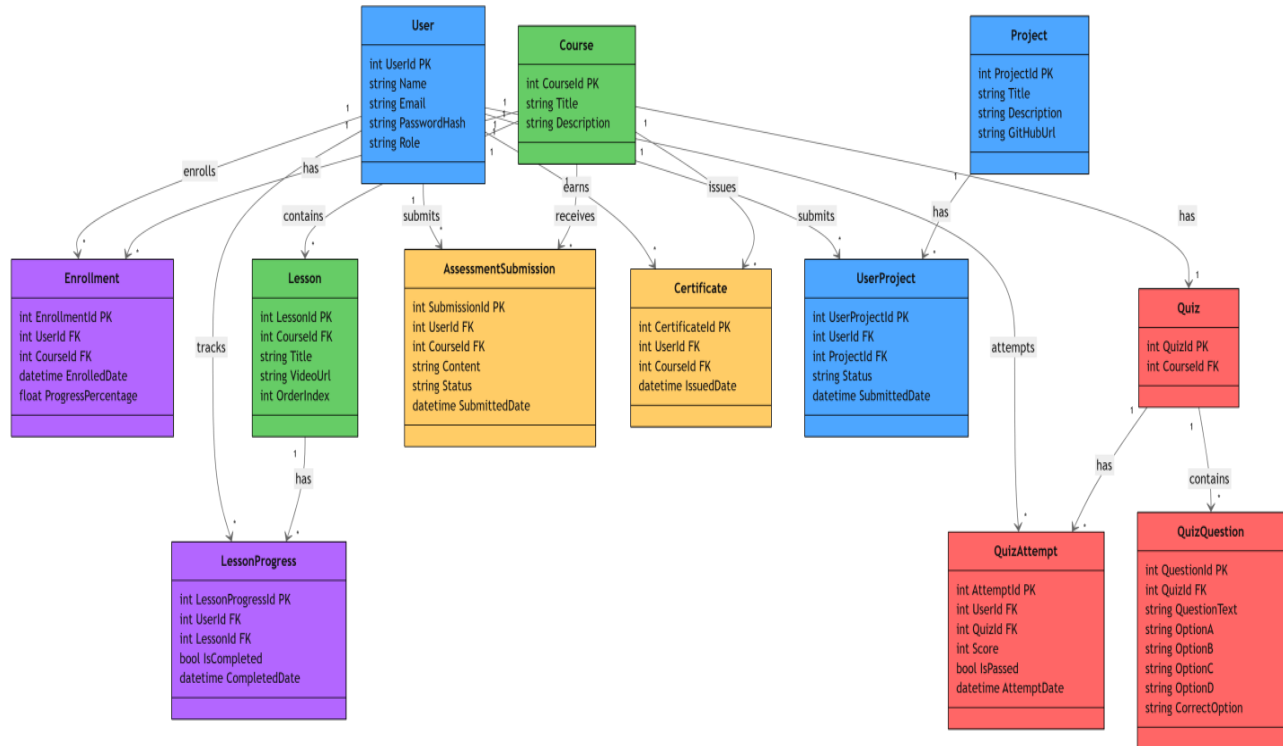
3.2.4.1 Authentication Flow



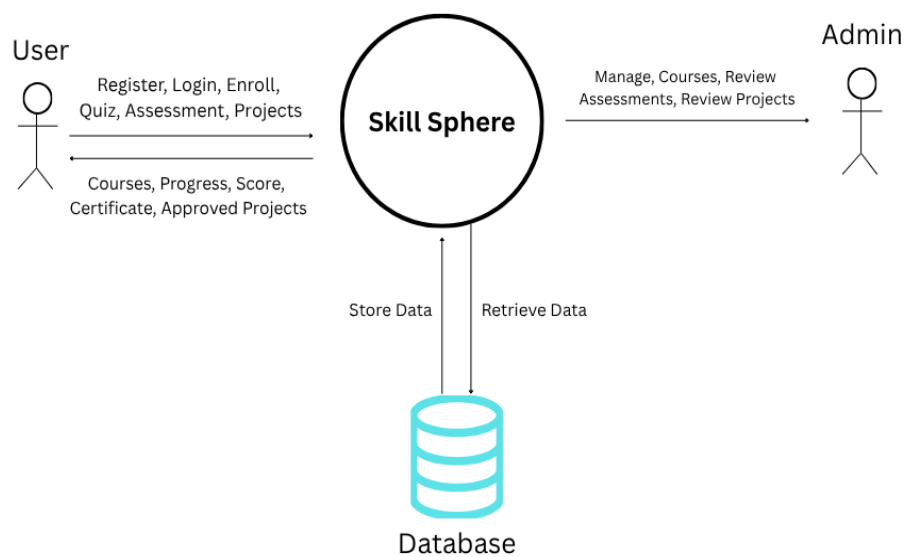
3.2.4.2 Certification Flow



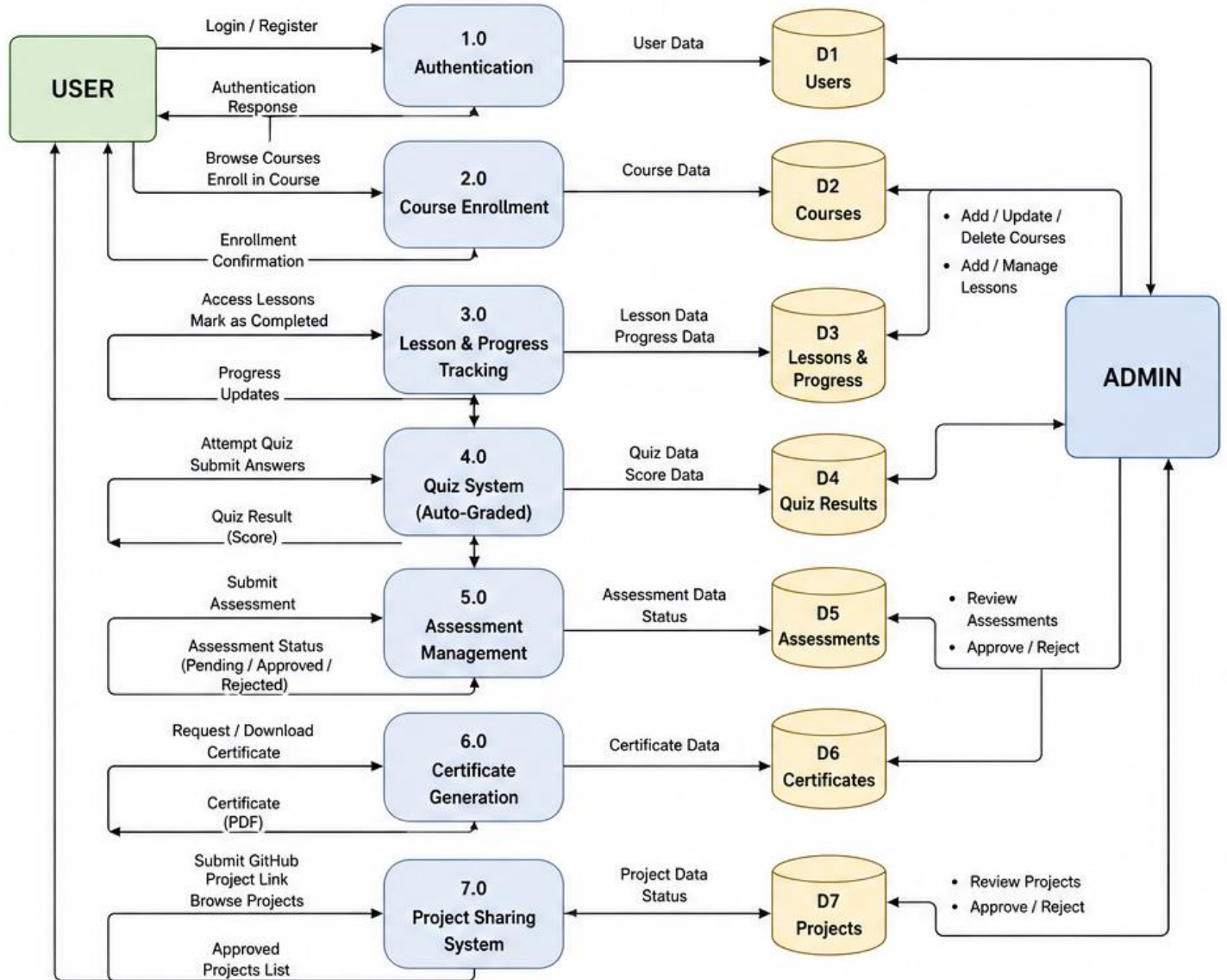
3.2.5 Class Diagram



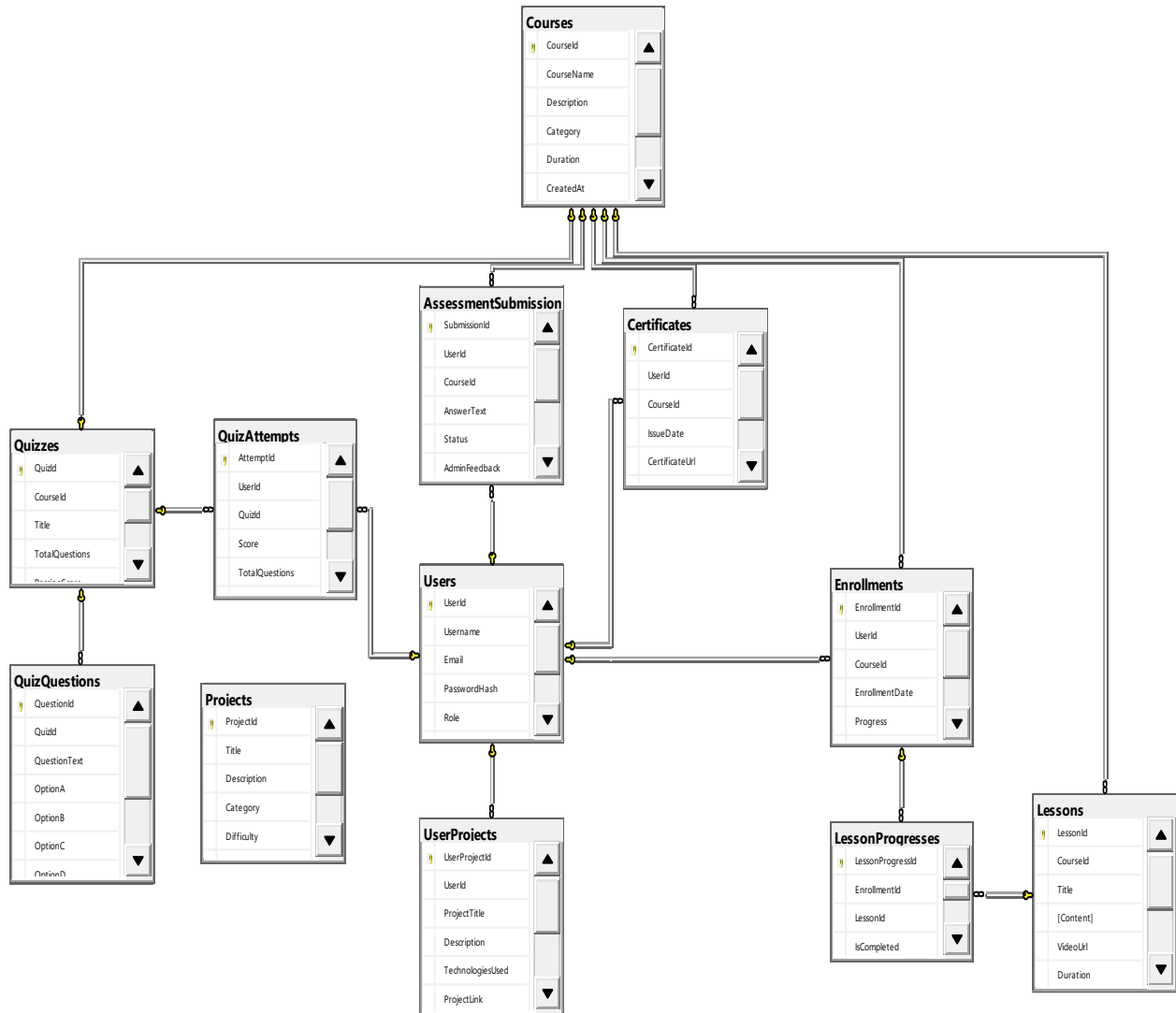
3.2.6 Data Flow Diagram (DFD Level 0)



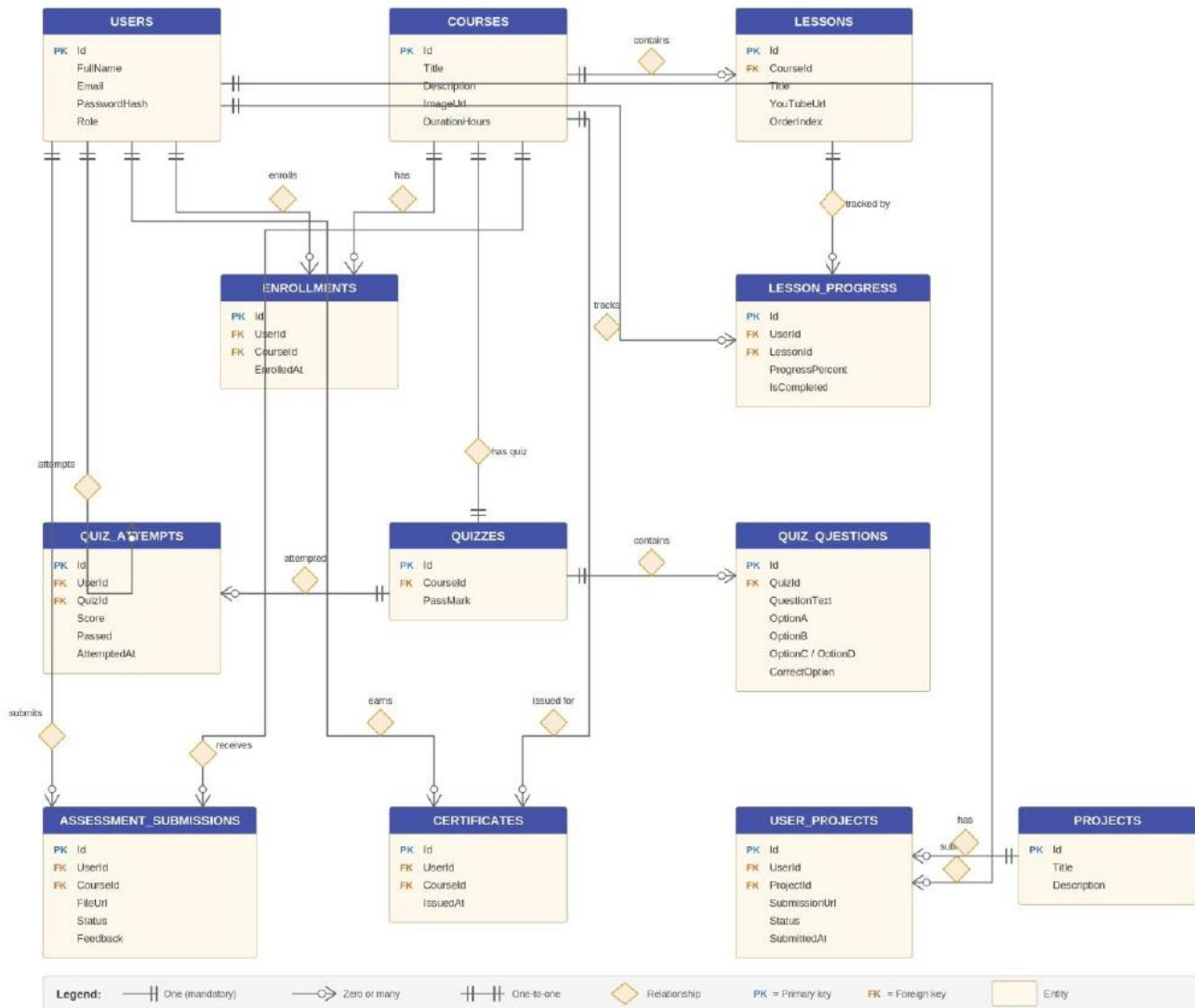
3.2.6 Data Flow Diagram (DFD Level 1)



Database Design



3.3.1 Entity Relationship Diagram (ERD)



3.4 System Architecture

Skill Sphere follows a three-tier client-server architecture consisting of the Presentation Layer, Application Layer, and Data Layer.

Presentation Layer (Frontend): The frontend is built using React with Vite as the build tool. It serves as the user interface layer that users and the admin interact with through a web browser. The frontend consists of modular React components organized into pages: Landing Page, Authentication Pages (Login/Register), User Dashboard, Admin Dashboard, Courses Page, Lesson Viewer with Side Panel, Quiz Page, Assessment Page, Certificate Page, Projects Page, and Upload Project Page. The frontend communicates with the backend exclusively through RESTful API calls using HTTP methods (GET,

POST, PUT, DELETE). The UI follows a teal and coral color theme for consistent visual identity.

Application Layer (Backend): The backend is built using C# with ASP.NET Core Web API. It serves as the business logic layer that processes all requests from the frontend, enforces business rules (such as quiz pass/fail logic, assessment approval workflow, and certificate generation conditions), and communicates with the database. The backend is organized into dedicated controllers: AuthController, AdminController, CoursesController, EnrollmentsController, LessonProgressController, QuizController, AssessmentController, CertificateController, ProjectsController, and UserProjectsController. Each controller handles specific API endpoints related to its domain. The backend implements password hashing, role-based authorization middleware, and input validation.

Data Layer (Database): The data layer consists of a SQL Server database (SkillSphereDB) hosted on a localdb instance, managed through SQL Server Management Studio (SSMS). The database stores all persistent data including user accounts, courses, lessons, enrollments, progress records, quiz questions, quiz attempts, assessment submissions, certificates, and project submissions. The data layer ensures data integrity through primary keys, foreign keys, unique constraints, and proper data types.

Communication Flow: User interacts with the React Frontend through a web browser. The Frontend sends HTTP requests to the ASP.NET Core Backend API. The Backend processes the request, applies business logic, and queries or updates the SQL Server Database. The Database returns the requested data to the Backend. The Backend formats the response and sends it back to the Frontend. The Frontend renders the updated data to the user.

3.5 System Workflow

The overall workflow of Skill Sphere can be summarized as follows:

The workflow begins when a new user visits the Skill Sphere platform and registers by providing their name, email, and password. The system validates the input, hashes the password, and stores the account in the database. The user then logs in using their credentials, and the system authenticates them and redirects them to the User Dashboard based on their role.

From the dashboard, the user browses available courses and enrolls in a course of their choice. The system creates an enrollment record and opens the course lesson page. The user watches video lessons one by one through the embedded YouTube player, with a side panel showing the lesson list and completion status. As the user completes each lesson, the system automatically tracks the progress and updates the completion percentage.

Once all lessons are completed, the quiz button becomes active. The user attempts the quiz consisting of 10 MCQs. The system auto-grades the quiz immediately upon

submission. If the user scores 7 or above, the quiz is marked as passed and the assessment submission becomes available. If the score is below 7, the user must reattempt the quiz until they pass.

After passing the quiz, the user submits a written assessment through the assessment form. The assessment is stored with a Pending status and appears in the admin panel for review. The admin logs in, views the pending submission, reads the assessment content, and either approves or rejects it. If rejected, the user may resubmit. If approved, the system automatically generates a digital certificate containing the user's name, course name, and completion status. The user can then download the certificate from the course page.

In parallel, users can submit GitHub project links through the project upload page. Each submission enters a Pending state. The admin reviews project submissions and approves or rejects them. Approved projects appear in the public Projects section where all users can browse and access them.

The admin also manages the platform by adding, updating, or deleting courses and lessons through the Admin Panel, ensuring that the course content remains current and relevant.

This workflow ensures a controlled, validated, and meaningful learning experience where every certification is backed by verified lesson completion, demonstrated knowledge through quizzes, and quality-assured assessment through admin review.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 Introduction to System Implementation

This chapter covers the implementation of Skill Sphere, including the development environment, tools used, backend and frontend development, database setup, and key features. The system follows a three-tier architecture with React on the frontend, ASP.NET Core C# Web API on the backend, and SQL Server as the database.

4.2 Development Environment and Tools

The following table lists all tools, technologies, and environments used during the development of Skill Sphere.

Category	Tool / Technology	Version
Frontend Framework	React (with Vite build tool)	React 18+
Backend Framework	ASP.NET Core Web API	C# / .NET 6+
Database	SQL Server (LocalDB)	SQL Server 2019+
DB Management	SQL Server Management Studio	SSMS 19+
Code Editor	Visual Studio Code	Latest
Backend IDE	Visual Studio	2022
Version Control	Git	Latest
API Testing	Postman / Swagger UI	Latest
Operating System	Windows 10 Pro	22H2
Hardware	Core i5, 8GB RAM, 256GB SSD	-
Package Manager	npm (frontend) / NuGet (backend)	Latest

4.3 Backend Implementation

The backend of Skill Sphere is built using C# with ASP.NET Core Web API. It serves as the application layer that handles all business logic, processes HTTP requests from the React frontend, enforces authorization rules, and communicates with the SQL Server database. The backend follows a controller-based architecture where each functional module of the system is handled by a dedicated controller.

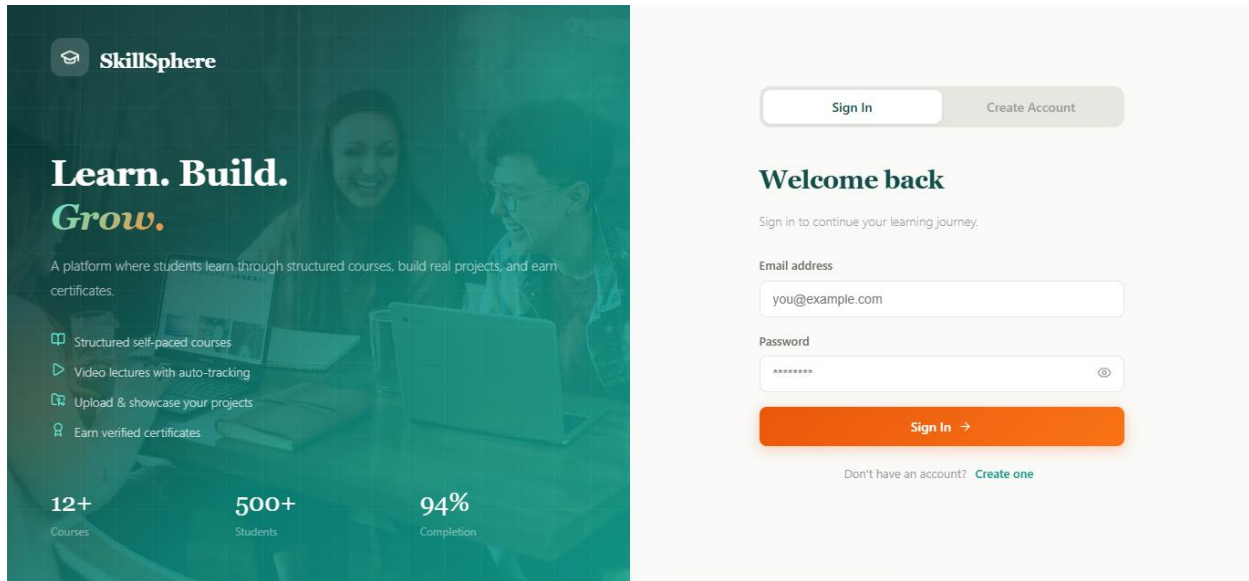
4.3.2 Database Connection Configuration

The backend connects to the SQL Server database using Entity Framework Core. The connection string is configured in the appsettings.json file, pointing to a LocalDB instance hosting the SkillSphereDB database. The DbContext class (SkillSphereDbContext) defines DbSet properties for each entity model, enabling the backend to perform CRUD operations on all database tables through Entity Framework.

4.3.4 Authentication and Authorization Implementation

The authentication system handles registration, login, and password reset. During registration, input fields are validated, duplicate emails are checked, passwords are hashed, and the user is stored with a default "User" role. The admin account is pre-seeded with an "Admin" role. During login, credentials are verified and the user is redirected to the User Dashboard or Admin Dashboard based on their role. Role-based access control restricts admin-only endpoints such as assessment approval, project approval, and course management.

Login Page



The login page features a teal banner on the left with the SkillSphere logo and the text "Learn. Build. Grow." Below this, it describes the platform as a place where students learn through structured courses, build real projects, and earn certificates. It lists four features: structured self-paced courses, video lectures with auto-tracking, upload & showcase your projects, and earn verified certificates. At the bottom of the banner, it shows statistics: 12+ Courses, 500+ Students, and 94% Completion. On the right, there are two buttons: "Sign In" and "Create Account". Below these is the heading "Welcome back" and a subheading "Sign in to continue your learning journey." The form includes fields for "Email address" (with the example "you@example.com") and "Password" (with a masked password "*****" and a toggle icon). A large orange "Sign In" button is at the bottom, with a link "Don't have an account? Create one" below it.

SkillSphere

Learn. Build. Grow.

A platform where students learn through structured courses, build real projects, and earn certificates.

- Structured self-paced courses
- Video lectures with auto-tracking
- Upload & showcase your projects
- Earn verified certificates

12+ Courses 500+ Students 94% Completion

[Sign In](#) [Create Account](#)

Welcome back

Sign in to continue your learning journey.

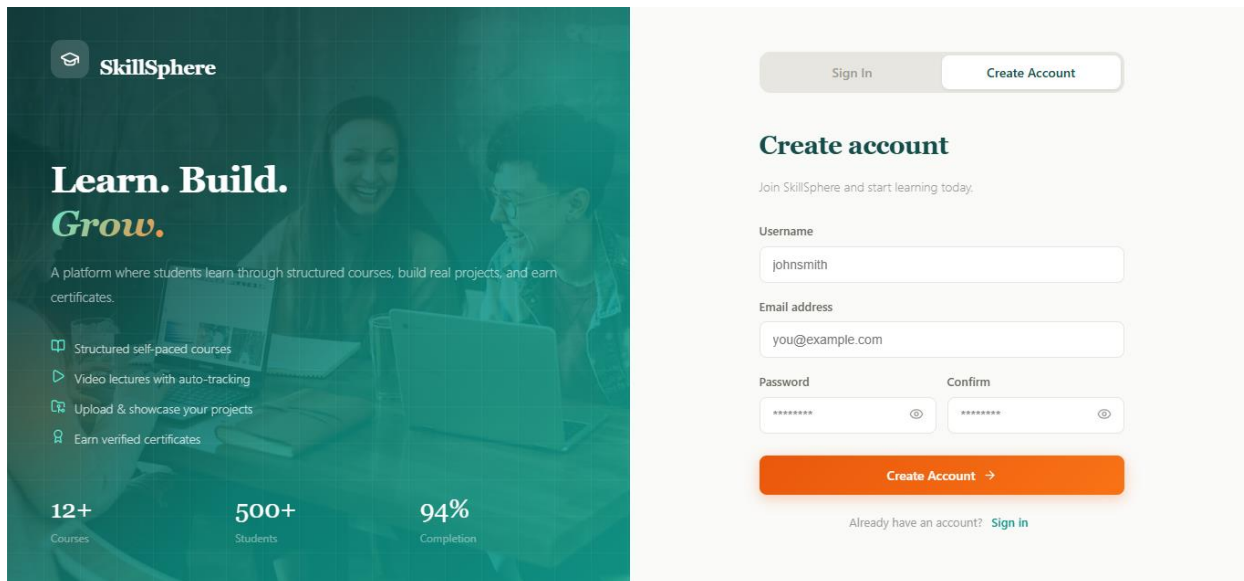
Email address
you@example.com

Password

[Sign In](#) →

Don't have an account? [Create one](#)

Registration Page



The registration page features the same teal banner as the login page. On the right, there are two buttons: "Sign In" and "Create Account". Below these is the heading "Create account" and a subheading "Join SkillSphere and start learning today." The form includes fields for "Username" (with the example "johnsmith"), "Email address" (with the example "you@example.com"), "Password" (with a masked password "*****" and a toggle icon), and "Confirm" (with a masked password "*****" and a toggle icon). A large orange "Create Account" button is at the bottom, with a link "Already have an account? Sign in" below it.

SkillSphere

Learn. Build. Grow.

A platform where students learn through structured courses, build real projects, and earn certificates.

- Structured self-paced courses
- Video lectures with auto-tracking
- Upload & showcase your projects
- Earn verified certificates

12+ Courses 500+ Students 94% Completion

[Sign In](#) [Create Account](#)

Create account

Join SkillSphere and start learning today.

Username
johnsmith

Email address
you@example.com

Password

Confirm

[Create Account](#) →

Already have an account? [Sign in](#)

4.3.5 Core Business Logic Implementation

The backend enforces several critical business rules that define the controlled learning progression of Skill Sphere:

Quiz Auto-Grading: When a user submits a quiz attempt, the QuizController receives the selected answers, compares them against the correct answers stored in the database,

calculates the score, and stores the attempt record with the score and pass/fail status. The passing threshold is hardcoded at 7 out of 10 correct answers.

Assessment Workflow: The `AssessmentController` manages the submission and review lifecycle. When a user submits an assessment, it is stored with a "Pending" status. The admin can then approve or reject the submission through dedicated endpoints, which update the status field in the database.

Certificate Generation: The `CertificateController` verifies that all three conditions are met before generating a certificate: all lessons in the course are completed (checked via `LessonProgress` records), the quiz is passed (checked via `QuizAttempt` records), and the assessment is approved (checked via `AssessmentSubmission` status). Only when all conditions are satisfied does the system create a certificate record and make it available for download.

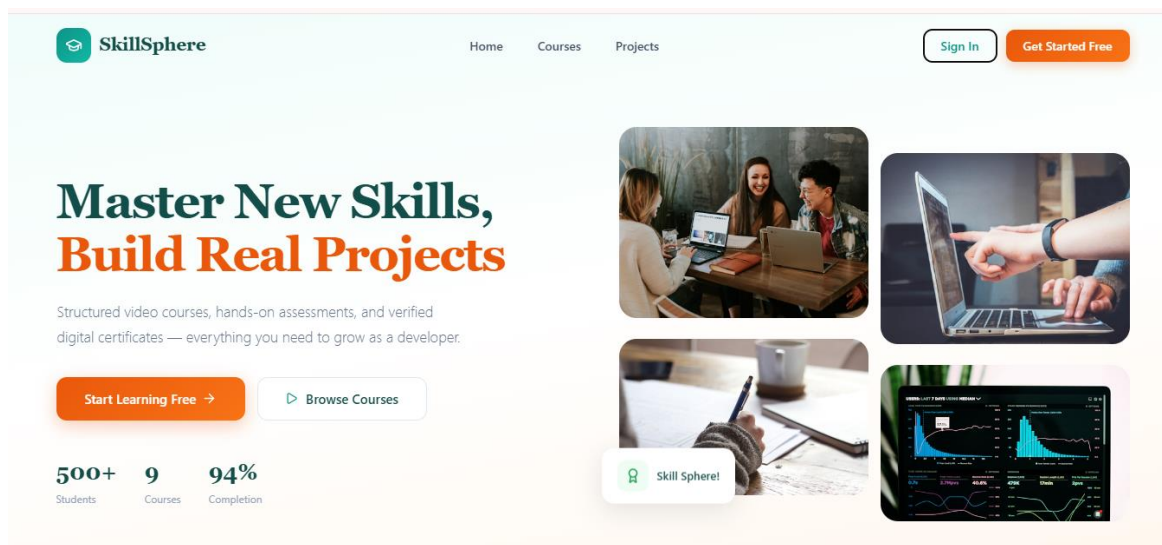
Project Approval: The `UserProjectsController` handles project link submissions. Each submission is stored with a "Pending" status. The admin reviews and approves or rejects submissions. Only approved projects are returned by the public projects endpoint.

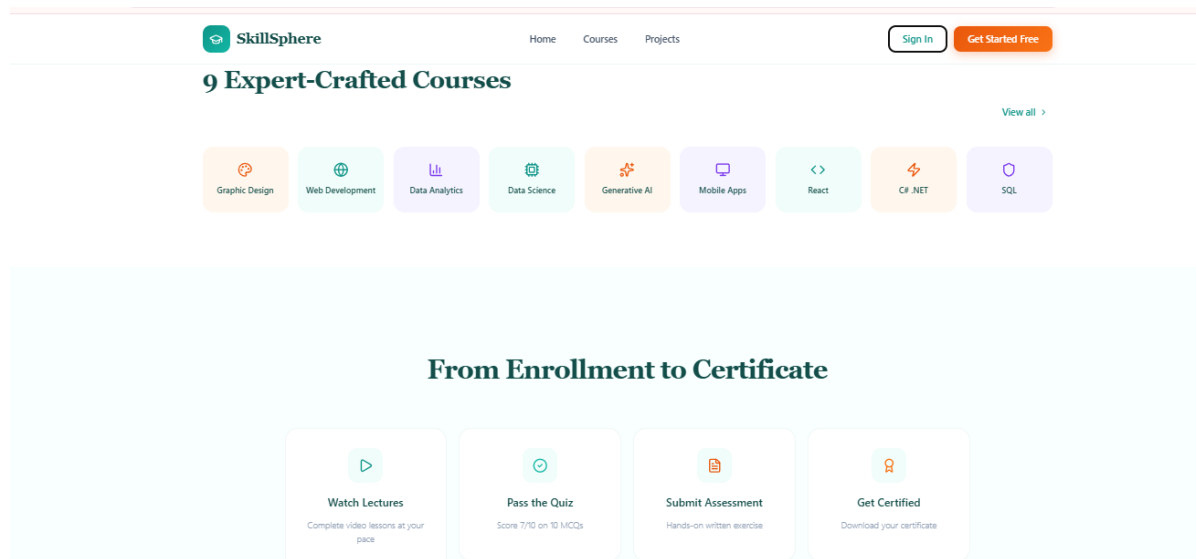
4.4 Frontend Implementation

The frontend of Skill Sphere is built using React with Vite as the build tool. It serves as the presentation layer that users and the admin interact with through a web browser. The frontend communicates with the backend exclusively through RESTful API calls using the `fetch` API or `Axios` for HTTP requests. The user interface follows a teal and coral color theme for consistent visual identity across all pages.

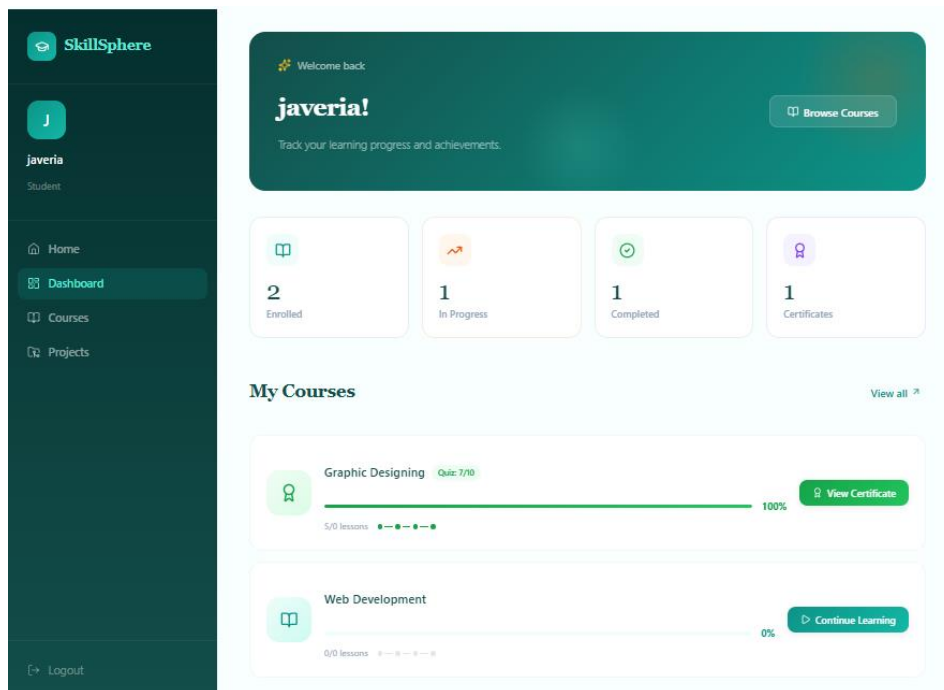
4.4.1 Key Pages and Their Implementation

Landing Page:

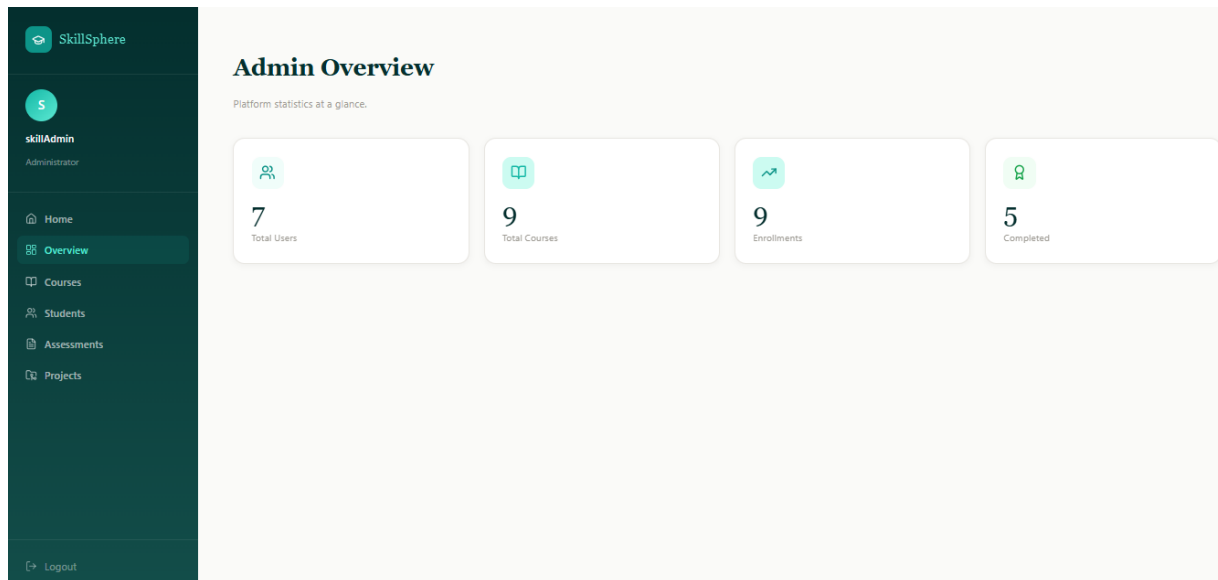




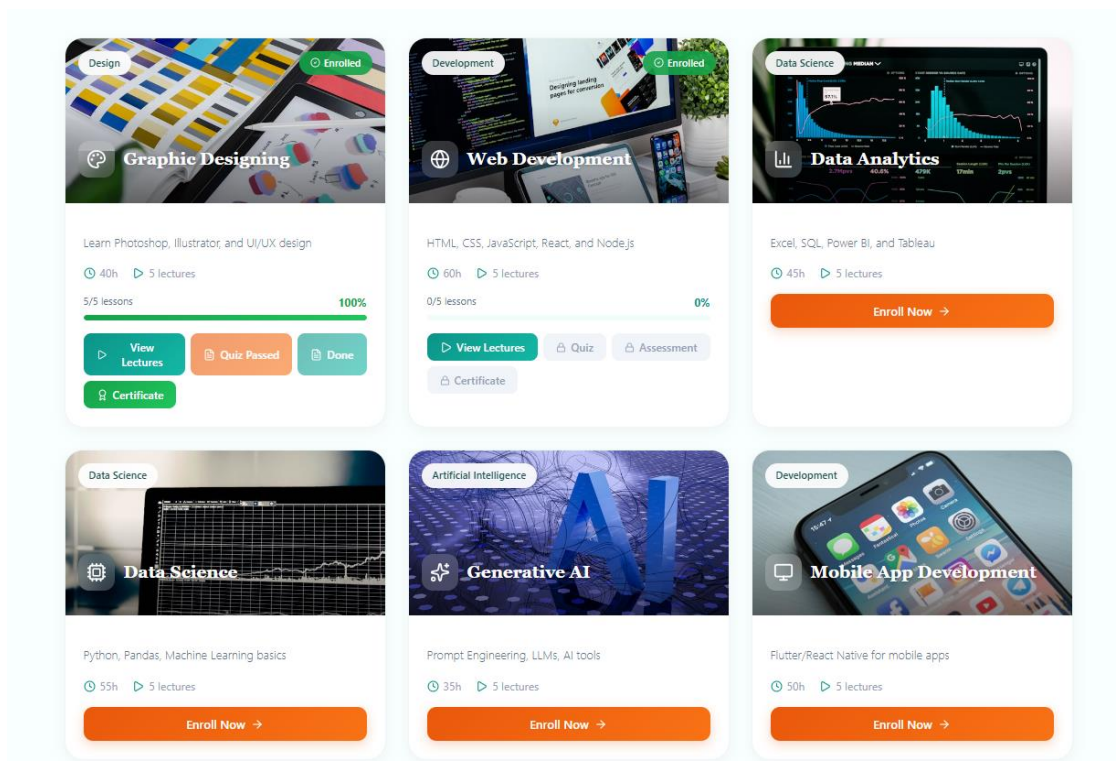
User Dashboard: After logging in, regular users are redirected to the User Dashboard. This page displays all courses the user is enrolled in, along with the progress percentage for each course. It serves as the central navigation hub from which users can access their enrolled courses, view their progress, and navigate to quizzes, assessments, and certificates.



Admin Dashboard: The admin is redirected to a separate Admin Dashboard upon login. This dashboard provides an overview of system statistics including total users, total courses, pending assessment submissions, and pending project submissions. It also provides navigation to course management, lesson management, assessment review, and project review sections.



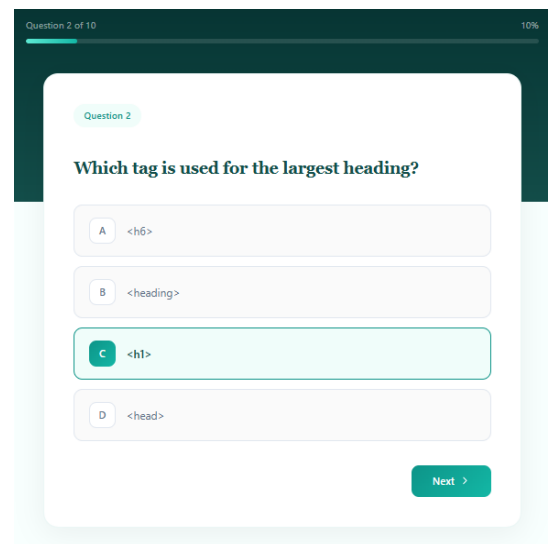
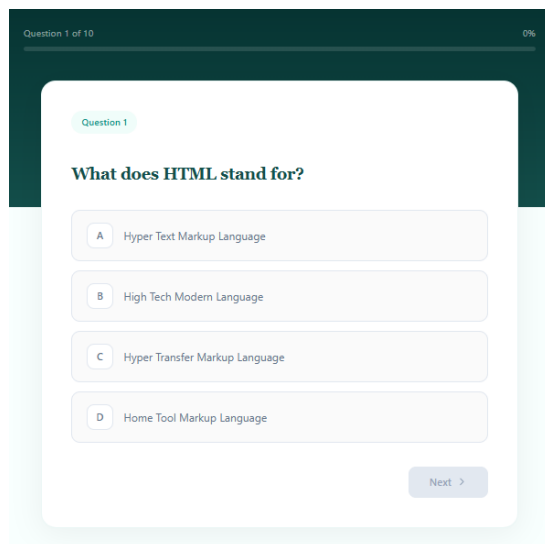
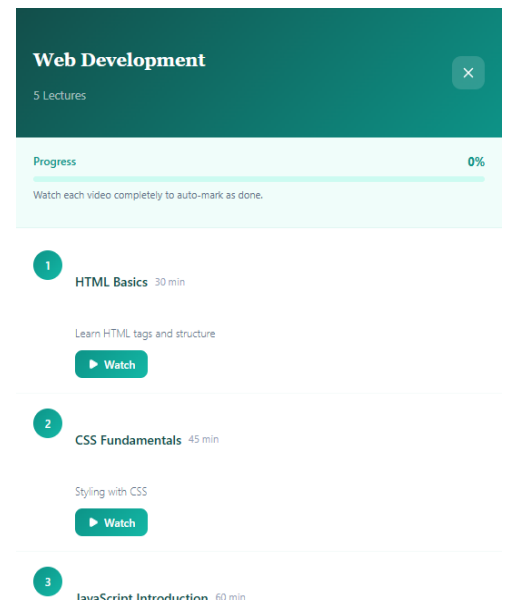
Courses Page: This page displays all available courses on the platform. Each course is presented as a card showing the course title and description. Users can click on a course to view its details and enroll if they are not already enrolled.



Course Lessons Page: When a user opens an enrolled course, the lessons page is displayed. This page features a side panel on the left listing all lessons in the course with their completion status (completed or pending). The main content area displays the embedded YouTube video player for the selected lesson. As the user watches each video,

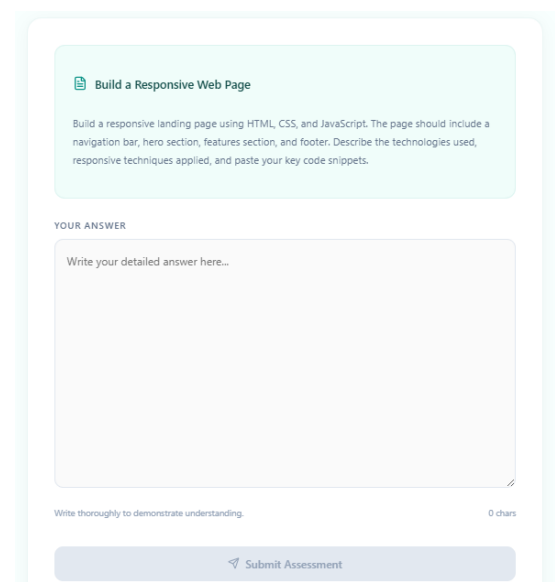
the system automatically tracks the completion and updates the progress bar. The quiz, assessment, and certificate buttons are always visible at the bottom or side of the page with appropriate lock/unlock states based on the user's progress.

Quiz Page: The quiz page loads 10 multiple-choice questions for the selected course. Each question displays four answer options. The user selects answers and submits the quiz. The system immediately displays the score and whether the user has passed or failed. If the user scores 7 or above, a success message is shown and the assessment becomes accessible.



Assessment Page:

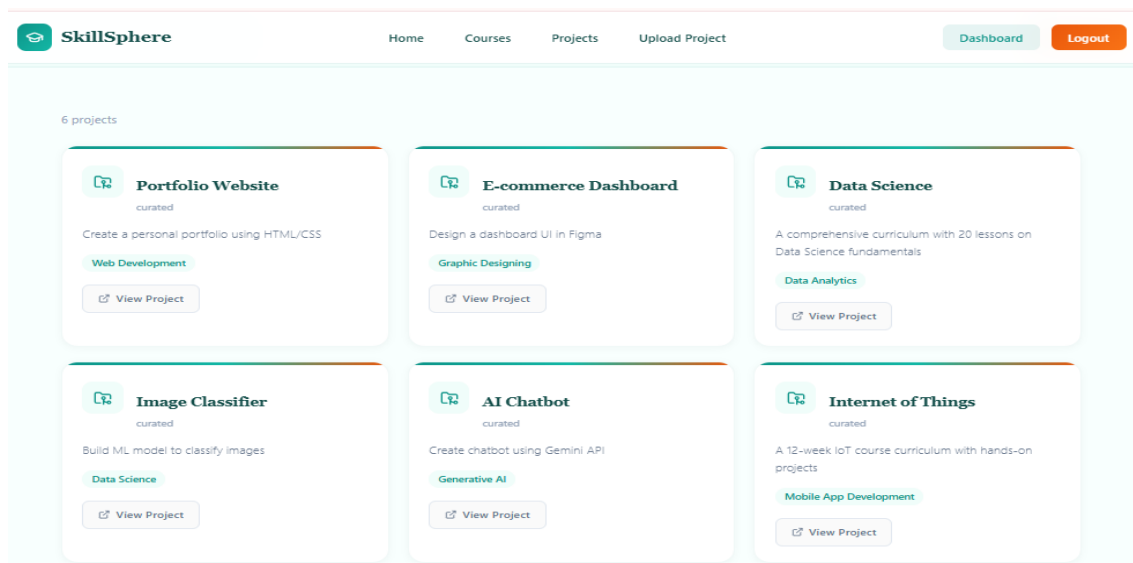
After passing the quiz, the user can access the assessment submission form. The page provides a text area where the user writes their assessment response. Upon submission, the system confirms that the assessment has been submitted and is pending admin review.



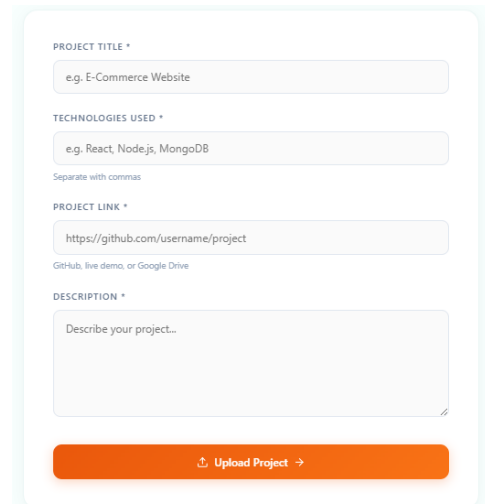
Certificate Page: Once the admin approves the assessment, the certificate becomes available. The certificate page displays the generated digital certificate containing the user's name, the course name, and the completion status. A download button allows the user to save the certificate as a file.



Projects Page: This page displays all approved projects submitted by users across the platform. Each project entry shows the project title, description, and a clickable GitHub link. Users can browse and discover projects shared by other learners.



Upload Project Page: Registered users can submit their own projects through this page. The form accepts a project title, description, and GitHub repository URL. The system validates that the URL is a valid GitHub link before accepting the submission.

A screenshot of a web form titled "Upload Project". The form is enclosed in a light blue rounded rectangle. It contains four input fields: "PROJECT TITLE *" with a placeholder "e.g. E-Commerce Website", "TECHNOLOGIES USED *" with a placeholder "e.g. React, Node.js, MongoDB" and a note "Separate with commas", "PROJECT LINK *" with a placeholder "https://github.com/username/project" and a note "GitHub, live demo, or Google Drive", and a "DESCRIPTION *" text area with a placeholder "Describe your project...". At the bottom is an orange button with a cloud upload icon and the text "Upload Project →".

4.5 Database Implementation

The database for Skill Sphere is implemented in SQL Server using a LocalDB instance. The database is named SkillSphereDB and is managed through SQL Server Management Studio (SSMS). The database schema consists of 12 tables that store all persistent data for the system. Entity Framework Core is used as the Object-Relational Mapping (ORM) tool to interact with the database from the backend.

4.6 Key Feature Implementation Details

4.6.1 Automatic Video Progress Tracking

Skill Sphere uses embedded YouTube videos for lesson delivery through iframe embeds. The system tracks lesson completion automatically. When a user opens a lesson and watches the video content, the frontend sends a completion request to the LessonProgressController endpoint. The backend creates a LessonProgress record marking that lesson as completed for the user's enrollment. The progress percentage displayed on the dashboard is calculated by dividing the number of completed lessons by the total number of lessons in the course and multiplying by 100.

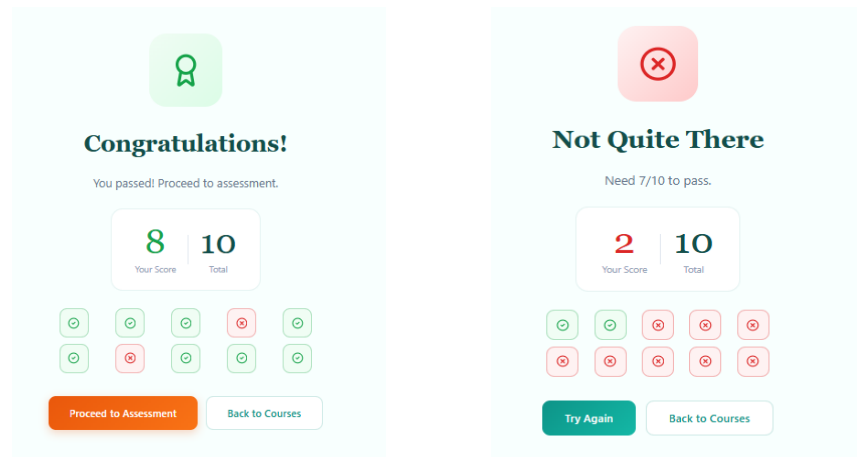
The YouTube videos are embedded using iframe tags with the video URL from the database. The system does not redirect users to YouTube; instead, the video plays directly within the lesson page, ensuring users remain on the platform throughout the learning experience.

4.6.2 Quiz Auto-Grading System

Each course has exactly 10 multiple-choice questions stored in the QuizQuestions table. When a user attempts a quiz, all 10 questions are loaded on the Quiz Page with four options each. The user selects one answer per question and submits the quiz. The frontend sends the selected answers to the QuizController, which compares each answer against the CorrectOption field in

the database, calculates the total score, and determines whether the user has passed (score of 7 or above) or failed.

The quiz attempt is stored in the QuizAttempts table with the score, pass/fail status, and timestamp. If the user passes, the assessment submission is unlocked on the frontend. If the user fails, they are informed that they must reattempt the quiz. The quiz button on the course page reflects the current state: locked if lessons are incomplete, available if lessons are complete but quiz is not yet passed, and completed if the quiz has been passed.



4.6.3 Digital Certificate Generation

Certificate generation is triggered when the admin approves a user's assessment. The CertificateController verifies three conditions before generating a certificate: all lessons completed, quiz passed (7/10 or above), and assessment approved. Once confirmed, a certificate record is created in the database and made available for download on the Certificate Page.

4.7 User Interface Design

Skill Sphere uses a teal and coral color theme across all pages. Teal serves as the primary color for navigation bars, buttons, and headings, while coral is used for call-to-action buttons and highlights. The navbar provides access to the dashboard, courses, projects, and account options on every page. The course lessons page includes a side panel for lesson navigation. Buttons display clear visual states: locked buttons appear greyed out, active buttons use teal or coral with hover effects, and completed items show a checkmark. All forms include real-time validation with error messages below input fields, and success actions are confirmed through on-screen messages.

4.8 Development and Deployment Configuration

Skill Sphere runs on localhost during development. The React frontend runs on Vite (port 5173), the ASP.NET Core backend runs on its own server (port 5187), and the database runs on a SQL Server LocalDB instance. CORS is configured in the backend to allow communication between the frontend and API endpoints

CHAPTER 6

TESTING AND EVALUATION

6.1 Testing Strategy

Skill Sphere was tested using a manual testing approach. The testing process included unit testing for individual modules, integration testing for module interaction, system testing for end-to-end workflows, and basic security testing for access control. All tests were executed in the local development environment using Postman for API testing and Chrome browser for frontend testing.

6.2 Unit Testing

Test ID	Module	Input	Expected Output	Result
UT-01	Registration	Valid credentials	Account created, redirected to login	Pass
UT-02	Registration	Duplicate email	"Email already exists" error	Pass
UT-03	Registration	Empty fields	Validation errors displayed	Pass
UT-04	Login	Valid user credentials	Redirected to User Dashboard	Pass
UT-05	Login	Invalid password	"Invalid credentials" error	Pass
UT-06	Login	Admin credentials	Redirected to Admin Dashboard	Pass
UT-07	Forgot Password	Registered email	Password updated successfully	Pass
UT-08	Forgot Password	Unregistered email	"Email not found" error	Pass
UT-09	Quiz	8/10 correct answers	Score: 8/10, Passed	Pass
UT-10	Quiz	5/10 correct answers	Score: 5/10, Failed	Pass
UT-11	Assessment	Valid submission	Status set to Pending	Pass
UT-12	Assessment	Empty submission	Validation error displayed	Pass
UT-13	Project Upload	Valid GitHub URL	Project submitted, status Pending	Pass
UT-14	Project Upload	Invalid URL	Validation error displayed	Pass
UT-15	Certificate	All conditions met	Certificate generated successfully	Pass

6.3 Integration Testing

Test ID	Modules Integrated	Expected Result	Result
IT-01	Frontend + Backend API	API responses rendered correctly on UI	Pass

IT-02	Backend + Database	Records created and retrieved correctly	Pass
IT-03	Enrollment + Lesson Progress	Progress tracked after enrollment	Pass
IT-04	Quiz + Assessment	Assessment unlocked after quiz passed	Pass
IT-05	Assessment + Certificate	Certificate unlocked after admin approval	Pass

6.4 System Testing

A complete end-to-end test was performed: a user registered, logged in, enrolled in a course, completed all 5 lessons, passed the quiz, submitted the assessment, received admin approval, and downloaded the certificate. The admin workflow was also tested including course management, assessment review, and project approval. All workflows executed successfully without errors.

Security Testing

Test ID	Test Scenario	Result
ST-01	Access dashboard without login	Redirected to login page - Pass
ST-02	Access admin panel as regular user	Access denied, redirected - Pass
ST-03	Password stored as plain text check	Passwords are hashed - Pass
ST-04	Duplicate registration attempt	Blocked with error message - Pass

6.6 User Acceptance Testing

The system was tested by 3 users who were given no prior instructions. Their feedback is summarized below.

Feedback	Action Taken
Navigation was clear and intuitive	No change needed
Quiz results displayed instantly	No change needed
Certificate download worked smoothly	No change needed
Side panel made lesson navigation easy	No change needed

6.7 Testing Summary

Metric	Value
Total Test Cases	25
Passed	25
Failed	0
Pass Rate	100%
Modules Tested	11
Testing Types	Unit, Integration, System, Security, UAT

All test cases passed successfully. The system meets all functional and non-functional requirements defined in the SRS and is ready for deployment.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

Skill Sphere is a web-based e-learning platform developed using React, ASP.NET Core C# Web API, and SQL Server. The system provides a controlled learning environment where users must complete lessons, pass a quiz (7/10), and receive admin approval on their assessment before earning a digital certificate. The platform also includes a project sharing feature with admin approval workflow. The system was designed, implemented, and tested successfully, meeting all functional and non-functional requirements defined in the SRS.

7.2 Limitations

The system currently runs on localhost and is not deployed to a live server. Only one admin account is supported. The forgot password feature resets the password without email verification. There is no email notification system for assessment approvals or rejections. The platform does not support mobile-responsive design optimization.

7.3 Future Enhancements

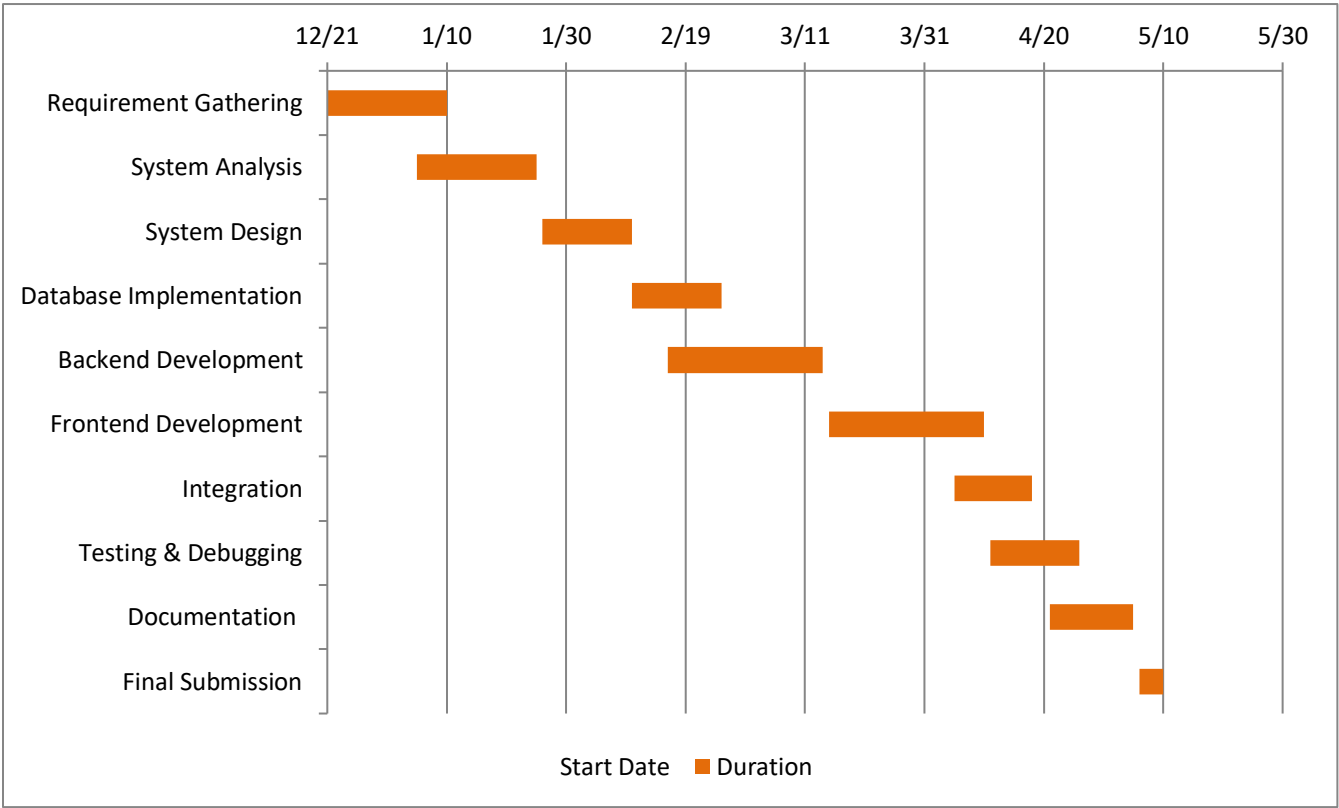
Future versions of Skill Sphere could include deployment on Microsoft Azure for public access, email notifications for assessment and project status updates, a mobile-responsive UI, multi-admin support, discussion forums for student interaction, payment integration for premium courses, and an AI-based course recommendation system.

7.4 Final Remarks

Skill Sphere successfully demonstrates a complete e-learning workflow from enrollment to certification. The three-tier architecture ensures clean separation of concerns, and the controlled learning progression enforces a structured educational experience. The project fulfills all requirements of the Final Year Project and is ready for future expansion.

I can see the IEEE citation format from your template. Let me write a bibliography for Skill Sphere with relevant references in the same format.

Gantt chart



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